

# Restoring M1/M8 and Eager/Compiled Agreement on RDNA4

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## Abstract

A pinned Radiance/vLLM inference stack produced different target logits when the same saved token history was processed one position at a time (M1) or eight positions at a time under D7 speculative verification (M8). The investigation localized numerical differences in GDN prefill and decode, convolution, gated normalization, attention, residual normalization and the BF16 vocabulary head. Four performance changes reduced a repaired verification round from approximately 76 ms to 60 ms in the measured 60K-input control.

A later eager/compiled investigation isolated intermediate BF16 rounding and native RoPE multiplication differences. Aligning both produced identical full logit vectors at 320 decode positions and the final prefill prediction, with identical captured inputs and outputs at all 465 matched activation boundaries. The combined repair is now measured directly as **eager M1 versus compiled M8** before and after both fixes, using the same 10,000-position Pi corpus.

This is an empirical result on one pinned configuration; it does not establish arbitrary-input equivalence or improved task accuracy.

The work consists of **two major fixes**:

| Revision             | What changed                                                                                                    | Comparison it repairs                    |
|----------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Original             | Pinned backend before these repairs                                                                             | Baseline                                 |
| Fix 1                | Align M1/M8 arithmetic and state transitions, then recover performance                                          | Compiled M8 versus compiled M1           |
| Final: Fix 1 + Fix 2 | Preserve BF16 intermediate rounding in compiled execution and use nearest-even RoPE products in eager execution | Compiled M8 versus eager M1 and eager M8 |

## 1. Configuration and reference

- One AMD Radeon AI PRO R9700, gfx1201, TP1.
- Radiance 1.0.16 / vLLM 0.28, PyTorch 2.12.0, ROCm 7.14, Triton 3.7.1.
- Qwen3.8-27B-Uncensored-MXFP4-awq; 64 layers, including 48 GDN and 16 attention layers; hidden width 5,120; vocabulary size 248,320.
- The target vocabulary head uses the full original BF16 weights in both arms. INT2 candidate selection is excluded from the equality claim. Other model weights, activation quantization and cache formats remain part of the pinned quantized backend; this is not an unquantized BF16-model comparison.
- The compiled comparisons use Inductor and piecewise GPU graphs, with capture sizes [1, 2, 4, 8]. Actual target graph replay is audited.
- The numerical reference is the explicitly selected serial implementation. Repairs also affect M1/preparation paths, so repaired M1 is measured afresh.

The 10K corpus comprises 23 saved Pi continuations with 57,008–65,527-token prefixes. Both arms consume identical saved token histories at aligned logical positions. Each M8 group processes eight positions with all seven supplied proposals accepted. The denominator is 10,000 processed decode positions, excluding intermediate-prefill head calls. Final initial-prefill predictions have a separate denominator of 23. No generated tools execute during replay.

## 2. Agreement definitions

For each position  $t$ , let  $z1(t)$  and  $z8(t)$  be the full vocabulary logit vectors.  $T\_k$  sorts logits by decreasing score and then increasing token ID for ties.

```

set_agreement(k) = count(set(T_k(z1)) == set(T_k(z8))) / N
order_agreement(k) = count(T_k(z1) == T_k(z8)) / N
mean_shared(k) = sum(|set(T_k(z1)) intersect set(T_k(z8))|) / N

```

Retained-score equality, inclusive boundary-tie membership and full-vector SHA-256 equality are separate checks. Matching hashes are strong empirical evidence, not a collision-free mathematical proof. Agreement measures execution consistency; it is not a percentage of questions answered correctly.

### 3. Top-1/10/20 and ordering results

#### Eager M1 versus compiled M8: before and after both fixes, full 10K corpus

**Before:** original eager M1 versus original compiled M8. **After both fixes:** final eager M1 versus final compiled M8 (Fix 1 + Fix 2). Each revision has a freshly measured eager M1 reference. All four runs use the same 23 continuations, context lengths and full BF16 target head. Compiled M8 uses Inductor and piecewise GPU graphs; eager M1 disables compilation and graphs.

| Prediction | Before: same set           | Before: same order         | Before: mean shared | After both fixes: same set | After both fixes: same order | After both fixes: mean shared |
|------------|----------------------------|----------------------------|---------------------|----------------------------|------------------------------|-------------------------------|
| Top 1      | 9,838 / 10,000<br>(98.38%) | 9,838 / 10,000<br>(98.38%) | 0.9838 / 1          | 10,000 / 10,000<br>(100%)  | 10,000 / 10,000<br>(100%)    | 1 / 1                         |
| Top 10     | 4,878 / 10,000<br>(48.78%) | 761 / 10,000<br>(7.61%)    | 9.3516 / 10         | 10,000 / 10,000<br>(100%)  | 10,000 / 10,000<br>(100%)    | 10 / 10                       |
| Top 20     | 2,313 / 10,000<br>(23.13%) | 4 / 10,000 (0.04%)         | 18.6883 / 20        | 10,000 / 10,000<br>(100%)  | 10,000 / 10,000<br>(100%)    | 20 / 20                       |

**Same set** means the same tokens regardless of order. **Same order** additionally requires identical ranking. **Mean shared** is the average number of shared tokens per position.

After both fixes, full-vocabulary hashes match at **10,000/10,000 decode positions** and **23/23 initial-prefill predictions**. Top-1/10/20 retained scores and inclusive boundary-tie sets also match throughout. [Counts, run identities and completion audit](#).

#### Fresh four-way comparison: 320 positions

Seven fresh native runs used the same saved 60,000-token Pi prefix and 320 aligned decode positions. Each M1/M8 pair uses its own revision's M1. Compiled arms ran with Inductor and piecewise GPU graphs; eager arms explicitly disabled compilation and graphs.

**Reading paired values: first = same token set (any order); second = same ranked order.** 320/320; 320/320 means both checks matched at all 320 tested positions.

| Compared implementations                            | Top-1 set/order  | Top-10 set/order | Top-20 set/order | Full vectors exact |
|-----------------------------------------------------|------------------|------------------|------------------|--------------------|
| Original compiled M8 vs original compiled M1        | 315/320; 315/320 | 157/320; 18/320  | 83/320; 1/320    | 0/320              |
| Fix 1 compiled M8 vs Fix 1 compiled M1              | 320/320; 320/320 | 320/320; 320/320 | 320/320; 320/320 | 320/320            |
| Fix 1 compiled M8 vs Fix 1 eager M8                 | 319/320; 319/320 | 146/320; 14/320  | 66/320; 0/320    | 0/320              |
| Final compiled M8 vs final eager M8 (Fix 1 + Fix 2) | 320/320; 320/320 | 320/320; 320/320 | 320/320; 320/320 | 320/320            |

These are **whole-model** results. The isolated stage comparisons in section 4 use common correct inputs to prevent upstream errors contaminating the stage being measured. [Run identities and complete counts](#).

### 4. Compiled stage timings and isolated correctness measurements

Both timing profiles use the same 60,000-input-token Pi fixture and the optimized compiled configuration: **enforce\_eager=false, Inductor, PIECEWISE GPU graphs, capture sizes [1, 2, 4, 8]**. The actual traces contain **65 target GPU graph launches in every captured round**, in both arms. Runtime receipts are recorded in [final-study-profile-audit.json](#). These are not eager timings. The full BF16 target head is used on both sides.

The fresh profiles contain one incomplete target-kernel inventory in each arm: original round 8 and final round 7 (one-based). The stage table uses the **six paired complete rounds, 1-6**, selected by inventory completeness rather than timing. Every recorded dispatch, including excluded rounds, remains in the evidence. [Profile audit](#).

All times are milliseconds per verification round: summed GPU dispatch durations for all layer instances, divided by six. The 256 projection dispatches per round are assigned to their four operations in each of 64 layers. The approximately 25 ms gate/up figure is the joint MLP projection across all 64 layers; its individual layer contributions follow.

The four correctness columns report **vocabulary top-20 set/order agreement on 320 positions**, with each layer instance given common correct inputs. Only one instance is substituted at a time before the native reference remainder produces logits. A position passes an aggregated stage only when every instance passes. Exact local outputs and state can reuse the already validated reference remainder; shared implementations across revisions are identified explicitly. These are not rankings of intermediate activation coordinates. The all-layer requirement makes these rates stricter than a single whole-model comparison.

“Performance” marks overhead removed from the initial repaired implementation; it does not imply that a stage is faster than the original unrepaired M8. A Fix 1 label records alignment to the selected serial arithmetic. Some original compiled M1/M8 stages already agree with each other on this sample; their actual agreement counts remain shown rather than being labelled as failures.

[Per-layer comparison counts](#) retain the individual results behind the aggregate table, including top-1/10/20 and full logit-vector agreement.

Original comparisons use the original M1 and M8 stage implementations; Fix 1 comparisons use the repaired implementations. Every column uses the same final-correct captured inputs. Thus these isolated rates diagnose individual stages; they are not the whole-model rates from section 3.

Compiled diagnostic replay disables graph replay to expose individual calls; its ordinary outputs must first match the graph-enabled release. **Diagnostic replay durations are excluded from this timing table**. The original and final columns below use fresh graph-enabled release profiles.

**Reading each top-20 pair: first = the same 20 tokens, in any order; second = those 20 tokens in exactly the same ranked order.** 320/320; 320/320 means both checks passed at all 320 tested positions. A position passes a stage only when every tested layer instance agrees.

| Compiled stage                               | Correctness fix?                           | Old compiled M8 | Final fixed compiled M8 ms | Change ms | Old compiled M8 vs old compiled M1 Top-20 set/order | Fix 1 compiled M8 vs Fix 1 compiled M1 Top-20 set/order | Fix 1 compiled M8 vs Fix 1 eager M8 Top-20 set/order | Final compiled M8 vs final eager M8 Top-20 set/order | Timing explanation                                                                                                                               |
|----------------------------------------------|--------------------------------------------|-----------------|----------------------------|-----------|-----------------------------------------------------|---------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Embedding + first input normalization        | Fix 1: normalization; embedding unchanged  | 0.004           | 0.008                      | +0.004    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Preserve the reference normalization rounding; the original embedding/norm fusion is indivisible.                                                |
| Layer input residual/ normalization          | Fix 1 + performance                        | 0.155           | 0.346                      | +0.191    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Preserve reduction and rounding; retain FP32 residual sums in registers.                                                                         |
| GDN input activation FP8 quantization        | No correctness repair                      | 0.127           | 0.127                      | 0.000     | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same quantization kernel and call count; the small timing delta is unassigned.                                                                   |
| GDN input projection                         | No correctness repair                      | 3.875           | 3.873                      | -0.002    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Unchanged MXFP4 projection arithmetic; timing differences are observations, not a projection optimization.                                       |
| GDN layout/copies and buffer initialization  | Performance only                           | 0.284           | 0.332                      | +0.047    | Within convolution/ recurrence cuts                 | Within convolution/ recurrence cuts                     | Within convolution/ recurrence cuts                  | Within convolution/ recurrence cuts                  | Preserve packed QKV views; remove split materializations and repacking. Remaining copies are included.                                           |
| GDN convolution                              | Fix 1 + performance                        | 0.494           | 0.226                      | -0.268    | 9/320; 0/320                                        | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Match serial product/accumulation order and rolling history; packed transport removes surrounding copies.                                        |
| GDN recurrence and gates                     | Fix 1                                      | 1.162           | 1.219                      | +0.057    | 13/320; 0/320                                       | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Match gate precision, reduction order, recurrent-state transition and output rounding.                                                           |
| GDN output gated normalization               | Fix 1 + performance                        | 0.138           | 0.097                      | -0.041    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Keep the serial row tile while processing independent rows concurrently; original fused constituents remain grouped.                             |
| GDN output activation FP8 quantization       | No correctness repair                      | 0.128           | 0.131                      | +0.003    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same quantization kernel and call count; the small timing delta is unassigned.                                                                   |
| GDN output projection                        | No correctness repair                      | 1.884           | 1.884                      | 0.000     | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Unchanged MXFP4 arithmetic; no causal speedup claimed.                                                                                           |
| Attention input activation FP8 quantization  | No correctness repair                      | 0.043           | 0.042                      | -0.001    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same quantization kernel and call count; timing cause is not isolated.                                                                           |
| Attention input projection                   | No correctness repair                      | 1.099           | 1.099                      | 0.000     | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Unchanged QKV MXFP4 projection; no causal speedup claimed.                                                                                       |
| Attention Q/K normalization, RoPE and layout | Fix 1: normalization; Fix 2: RoPE rounding | 0.079           | 0.236                      | +0.157    | 320/320; 320/320                                    | 320/320; 320/320                                        | 14/320; 0/320                                        | 320/320; 320/320                                     | Preserve serial normalization and BF16 RoPE product rounding. Fused original constituents share one timing.                                      |
| Attention KV write                           | No correctness repair                      | 0.046           | 0.047                      | +0.001    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same cache-write kernel and KV format; timing cause is not isolated.                                                                             |
| Attention decode                             | Fix 1 + performance                        | 4.224           | 5.285                      | +1.061    | 22/320; 0/320 †                                     | 320/320; 320/320 †                                      | 320/320; 320/320 †                                   | 320/320; 320/320 †                                   | Preserve each query's causal tile/softmax decisions. Share KV reads within one or two tile-aligned query groups; two groups repeat context work. |
| Attention split-KV merge                     | Fix 1                                      | 0.138           | 0.120                      | -0.018    | 22/320; 0/320 †                                     | 320/320; 320/320 †                                      | 320/320; 320/320 †                                   | 320/320; 320/320 †                                   | Use each query's serial split/merge arithmetic; the observed reduction has not been isolated from the decode change.                             |
| Attention output gating                      | Fix 2: intermediate rounding               | 0.028           | 0.032                      | +0.004    | 320/320; 320/320                                    | 320/320; 320/320                                        | 20/320; 0/320                                        | 320/320; 320/320                                     | Preserve the BF16 sigmoid result before multiplying the output gate.                                                                             |
| Attention output activation FP8 quantization | No correctness repair                      | 0.046           | 0.046                      | 0.000     | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same quantization kernel and call count; timing cause is not isolated.                                                                           |
| Attention output projection                  | No correctness repair                      | 0.557           | 0.521                      | -0.035    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Unchanged output MXFP4 projection; no causal speedup claimed.                                                                                    |
| Post-attention/GDN residual/ normalization   | Fix 1 + performance                        | 0.142           | 0.353                      | +0.210    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Preserve serial reduction/rounding and keep residual values in registers.                                                                        |
| MLP gate/up input FP8 quantization           | No correctness repair                      | 0.165           | 0.167                      | +0.002    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same quantization kernel and call count; timing cause is not isolated.                                                                           |
| MLP gate/up projection                       | No correctness repair                      | 25.271          | 25.437                     | +0.166    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | One joint gate/up GEMM per layer, 64 per round. The per-layer table subdivides this total; gate and up have no separate measured durations.      |
| MLP SiLU and gating                          | Fix 2: intermediate rounding               | 0.156           | 0.189                      | +0.033    | 320/320; 320/320                                    | 320/320; 320/320                                        | 15/320; 0/320                                        | 320/320; 320/320                                     | Preserve the BF16 SiLU result before multiplication; retain one fused pointwise launch.                                                          |
| MLP down input FP8 quantization              | No correctness repair                      | 0.297           | 0.297                      | +0.001    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Same quantization kernel and call count; timing cause is not isolated.                                                                           |
| MLP down projection                          | No correctness repair                      | 5.147           | 5.143                      | -0.004    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Unchanged MXFP4 down projection; no causal speedup claimed.                                                                                      |
| Final normalization/layout                   | Fix 1                                      | 0.002           | 0.006                      | +0.003    | 320/320; 320/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Preserve final-normalization reduction and the BF16 rounding of its retained residual sum; isolated check includes that fused addition.          |
| Full BF16 target head                        | Fix 1 + performance                        | 4.024           | 4.140                      | +0.116    | 320/320; 319/320                                    | 320/320; 320/320                                        | 320/320; 320/320                                     | 320/320; 320/320                                     | Interleave two arithmetic-preserving M4 groups in one HIP launch, removing duplicated launch and concatenation overhead.                         |
| Drafter                                      | No target correctness repair               | 6.328           | 6.518                      | +0.190    | N/A: not a target prediction stage                  | N/A: not a target prediction stage                      | N/A: not a target prediction stage                   | N/A: not a target prediction stage                   | Unchanged proposal model; its timings and predictions are not target-M1 equivalence measurements.                                                |
| Other GPU bookkeeping                        | Not an isolated numerical stage            | 0.551           | 0.560                      | +0.009    | N/A: not a target prediction stage                  | N/A: not a target prediction stage                      | N/A: not a target prediction stage                   | N/A: not a target prediction stage                   | Sampling/state bookkeeping outside the model scopes. Exact semantic attribution is unavailable; each kernel remains listed below.                |

† Attention decode and split-KV merge have separate GPU timings but one native numerical interface, so their correctness cells report the joint comparison. GDN transport copies are included in the convolution/recurrence state checks; they do not have an independent vocabulary-prediction result. Drafter and bookkeeping work are outside the target-stage comparison.

Target-body subtotal: **45.693 ms old; 47.263 ms fixed**. Total recorded GPU kernel duration: **56.596 ms old; 58.481 ms fixed**. These are subtotals, not extra stages. Kernel durations can overlap and exclude host/queue gaps; they are not wall-clock round latency.

## Interpreting the timing differences

The FP8 quantizer is unchanged and runs 256 times per round in both versions. Its fresh totals are **0.806 ms original and 0.810 ms final**. The earlier large quantization timing difference does not reproduce in this comparison.

Attention decode averages **4.224 ms original and 5.285 ms final**. Final rounds split into approximately 3.55 ms and 7.02 ms groups. The repaired kernel uses one or two groups of queries aligned to 16-token tiles to preserve each query's M1 causal and softmax decisions; a second group repeats context work. The split-KV merge is timed separately: **0.138 ms original and 0.120 ms final**.

An unchanged stage's timing delta alone does not establish a performance fix.

## Every layer, including the 64 contributions to the gate/up total

Layer numbers are zero-based. "Other three projections" means the input, attention/GDN output, and MLP down projections; all individual semantic stage values are also retained in `stage-times.json`. Non-projection work includes normalization, gating, quantization, attention/GDN operations and copies.

Expand all 64 layers

| Layer (zero-based) | Type      | All layer work old ms | Final fixed ms | Gate/up old ms | Final fixed ms | Other three projections old ms | Final fixed ms | Non-projection old ms | Final fixed ms |
|--------------------|-----------|-----------------------|----------------|----------------|----------------|--------------------------------|----------------|-----------------------|----------------|
| 0                  | GDN       | 0.6255                | 0.6298         | 0.3712         | 0.3704         | 0.1946                         | 0.1945         | 0.0597                | 0.0649         |
| 1                  | GDN       | 0.6274                | 0.6246         | 0.3712         | 0.3664         | 0.1986                         | 0.1983         | 0.0577                | 0.0599         |
| 2                  | GDN       | 0.6321                | 0.6298         | 0.3716         | 0.3692         | 0.1989                         | 0.1986         | 0.0616                | 0.0620         |
| 3                  | Attention | 0.8453                | 0.9203         | 0.3747         | 0.3743         | 0.1831                         | 0.1801         | 0.2876                | 0.3659         |
| 4                  | GDN       | 0.6372                | 0.6387         | 0.3797         | 0.3784         | 0.1993                         | 0.1986         | 0.0582                | 0.0617         |
| 5                  | GDN       | 0.6423                | 0.6460         | 0.3828         | 0.3844         | 0.1996                         | 0.1989         | 0.0599                | 0.0627         |
| 6                  | GDN       | 0.6501                | 0.6501         | 0.3868         | 0.3858         | 0.2003                         | 0.1995         | 0.0630                | 0.0647         |
| 7                  | Attention | 0.8685                | 0.9506         | 0.3884         | 0.3909         | 0.1833                         | 0.1807         | 0.2968                | 0.3790         |
| 8                  | GDN       | 0.6468                | 0.6532         | 0.3876         | 0.3905         | 0.1991                         | 0.1987         | 0.0600                | 0.0641         |
| 9                  | GDN       | 0.6481                | 0.6572         | 0.3875         | 0.3924         | 0.2002                         | 0.2004         | 0.0604                | 0.0644         |
| 10                 | GDN       | 0.6543                | 0.6575         | 0.3902         | 0.3913         | 0.2000                         | 0.2008         | 0.0641                | 0.0653         |
| 11                 | Attention | 0.8717                | 0.9490         | 0.3912         | 0.3899         | 0.1826                         | 0.1802         | 0.2979                | 0.3788         |
| 12                 | GDN       | 0.6526                | 0.6541         | 0.3912         | 0.3905         | 0.1999                         | 0.2002         | 0.0615                | 0.0634         |
| 13                 | GDN       | 0.6521                | 0.6551         | 0.3904         | 0.3897         | 0.2005                         | 0.2009         | 0.0612                | 0.0645         |
| 14                 | GDN       | 0.6536                | 0.6587         | 0.3881         | 0.3939         | 0.2012                         | 0.1995         | 0.0643                | 0.0653         |
| 15                 | Attention | 0.8739                | 0.9551         | 0.3909         | 0.3940         | 0.1839                         | 0.1808         | 0.2991                | 0.3803         |
| 16                 | GDN       | 0.6519                | 0.6576         | 0.3909         | 0.3931         | 0.1998                         | 0.1999         | 0.0612                | 0.0646         |
| 17                 | GDN       | 0.6531                | 0.6607         | 0.3911         | 0.3952         | 0.2009                         | 0.2004         | 0.0610                | 0.0651         |
| 18                 | GDN       | 0.6587                | 0.6637         | 0.3910         | 0.3972         | 0.2016                         | 0.2004         | 0.0662                | 0.0661         |
| 19                 | Attention | 0.8778                | 0.9606         | 0.3926         | 0.3961         | 0.1831                         | 0.1815         | 0.3020                | 0.3831         |
| 20                 | GDN       | 0.6580                | 0.6628         | 0.3955         | 0.3949         | 0.2005                         | 0.2006         | 0.0620                | 0.0673         |

| Layer<br>(zero-based) | Type      | All layer<br>work old<br>ms | Final<br>fixed ms | Gate/up<br>old ms | Final<br>fixed ms | Other three<br>projections old ms | Final<br>fixed ms | Non-<br>projection old<br>ms | Final<br>fixed ms |
|-----------------------|-----------|-----------------------------|-------------------|-------------------|-------------------|-----------------------------------|-------------------|------------------------------|-------------------|
| 21                    | GDN       | 0.6583                      | 0.6637            | 0.3971            | 0.3955            | 0.1999                            | 0.2021            | 0.0614                       | 0.0660            |
| 22                    | GDN       | 0.6630                      | 0.6630            | 0.3977            | 0.3966            | 0.1999                            | 0.2009            | 0.0654                       | 0.0654            |
| 23                    | Attention | 0.8782                      | 0.9618            | 0.3938            | 0.3966            | 0.1831                            | 0.1812            | 0.3014                       | 0.3839            |
| 24                    | GDN       | 0.6550                      | 0.6633            | 0.3926            | 0.3970            | 0.2005                            | 0.2009            | 0.0619                       | 0.0653            |
| 25                    | GDN       | 0.6546                      | 0.6629            | 0.3926            | 0.3959            | 0.2005                            | 0.2013            | 0.0615                       | 0.0658            |
| 26                    | GDN       | 0.6622                      | 0.6633            | 0.3948            | 0.3968            | 0.2016                            | 0.2012            | 0.0659                       | 0.0653            |
| 27                    | Attention | 0.8835                      | 0.9642            | 0.3948            | 0.3978            | 0.1847                            | 0.1814            | 0.3040                       | 0.3850            |
| 28                    | GDN       | 0.6579                      | 0.6658            | 0.3946            | 0.3989            | 0.2016                            | 0.1999            | 0.0617                       | 0.0670            |
| 29                    | GDN       | 0.6606                      | 0.6682            | 0.3970            | 0.4009            | 0.2007                            | 0.2013            | 0.0628                       | 0.0661            |
| 30                    | GDN       | 0.6652                      | 0.6682            | 0.3983            | 0.4019            | 0.2010                            | 0.2005            | 0.0660                       | 0.0659            |
| 31                    | Attention | 0.8834                      | 0.9707            | 0.3966            | 0.4007            | 0.1844                            | 0.1811            | 0.3025                       | 0.3889            |
| 32                    | GDN       | 0.6588                      | 0.6671            | 0.3970            | 0.3999            | 0.2010                            | 0.2004            | 0.0608                       | 0.0668            |
| 33                    | GDN       | 0.6600                      | 0.6675            | 0.3965            | 0.4013            | 0.2013                            | 0.2005            | 0.0622                       | 0.0656            |
| 34                    | GDN       | 0.6646                      | 0.6704            | 0.3981            | 0.4022            | 0.2009                            | 0.2008            | 0.0656                       | 0.0674            |
| 35                    | Attention | 0.8900                      | 0.9715            | 0.4003            | 0.4021            | 0.1836                            | 0.1815            | 0.3062                       | 0.3880            |
| 36                    | GDN       | 0.6646                      | 0.6680            | 0.4001            | 0.4024            | 0.2017                            | 0.1997            | 0.0628                       | 0.0660            |
| 37                    | GDN       | 0.6616                      | 0.6701            | 0.3983            | 0.4033            | 0.2010                            | 0.2013            | 0.0622                       | 0.0655            |
| 38                    | GDN       | 0.6671                      | 0.6717            | 0.4007            | 0.4043            | 0.2011                            | 0.2009            | 0.0654                       | 0.0665            |
| 39                    | Attention | 0.8897                      | 0.9784            | 0.4000            | 0.4058            | 0.1831                            | 0.1815            | 0.3066                       | 0.3911            |
| 40                    | GDN       | 0.6629                      | 0.6723            | 0.3994            | 0.4064            | 0.2007                            | 0.2006            | 0.0627                       | 0.0652            |
| 41                    | GDN       | 0.6629                      | 0.6730            | 0.3993            | 0.4057            | 0.2008                            | 0.2006            | 0.0627                       | 0.0666            |
| 42                    | GDN       | 0.6651                      | 0.6751            | 0.3992            | 0.4062            | 0.2005                            | 0.2028            | 0.0655                       | 0.0661            |
| 43                    | Attention | 0.8903                      | 0.9743            | 0.4013            | 0.4038            | 0.1836                            | 0.1814            | 0.3053                       | 0.3892            |
| 44                    | GDN       | 0.6671                      | 0.6686            | 0.4023            | 0.4022            | 0.2015                            | 0.1999            | 0.0632                       | 0.0665            |
| 45                    | GDN       | 0.6669                      | 0.6672            | 0.4030            | 0.4014            | 0.2007                            | 0.2005            | 0.0632                       | 0.0653            |
| 46                    | GDN       | 0.6671                      | 0.6699            | 0.4009            | 0.4026            | 0.2004                            | 0.2012            | 0.0658                       | 0.0661            |
| 47                    | Attention | 0.8922                      | 0.9736            | 0.4026            | 0.4041            | 0.1834                            | 0.1819            | 0.3062                       | 0.3877            |
| 48                    | GDN       | 0.6687                      | 0.6745            | 0.4040            | 0.4057            | 0.2018                            | 0.2009            | 0.0628                       | 0.0679            |
| 49                    | GDN       | 0.6713                      | 0.6746            | 0.4064            | 0.4075            | 0.2016                            | 0.2004            | 0.0633                       | 0.0667            |
| 50                    | GDN       | 0.6709                      | 0.6721            | 0.4037            | 0.4043            | 0.2009                            | 0.2011            | 0.0663                       | 0.0667            |
| 51                    | Attention | 0.8912                      | 0.9744            | 0.4003            | 0.4037            | 0.1844                            | 0.1810            | 0.3065                       | 0.3898            |
| 52                    | GDN       | 0.6629                      | 0.6704            | 0.3994            | 0.4040            | 0.2005                            | 0.2001            | 0.0630                       | 0.0663            |
| 53                    | GDN       | 0.6618                      | 0.6711            | 0.3997            | 0.4047            | 0.2004                            | 0.2008            | 0.0617                       | 0.0656            |
| 54                    | GDN       | 0.6692                      | 0.6741            | 0.4012            | 0.4058            | 0.2018                            | 0.2018            | 0.0663                       | 0.0665            |
| 55                    | Attention | 0.8885                      | 0.9742            | 0.3995            | 0.4032            | 0.1838                            | 0.1811            | 0.3051                       | 0.3899            |
| 56                    | GDN       | 0.6586                      | 0.6692            | 0.3974            | 0.4028            | 0.1993                            | 0.2004            | 0.0619                       | 0.0660            |
| 57                    | GDN       | 0.6636                      | 0.6692            | 0.3996            | 0.4014            | 0.2012                            | 0.2019            | 0.0628                       | 0.0659            |
| 58                    | GDN       | 0.6692                      | 0.6695            | 0.4010            | 0.4021            | 0.2019                            | 0.2012            | 0.0664                       | 0.0662            |

| Layer (zero-based) | Type      | All layer work old ms | Final fixed ms | Gate/up old ms | Final fixed ms | Other three projections old ms | Final fixed ms | Non-projection old ms | Final fixed ms |
|--------------------|-----------|-----------------------|----------------|----------------|----------------|--------------------------------|----------------|-----------------------|----------------|
| 59                 | Attention | 0.8904                | 0.9620         | 0.4009         | 0.4019         | 0.1837                         | 0.1820         | 0.3059                | 0.3781         |
| 60                 | GDN       | 0.6652                | 0.6723         | 0.4008         | 0.4053         | 0.2015                         | 0.2004         | 0.0630                | 0.0667         |
| 61                 | GDN       | 0.6649                | 0.6802         | 0.4011         | 0.4099         | 0.2011                         | 0.2021         | 0.0627                | 0.0682         |
| 62                 | GDN       | 0.6688                | 0.6779         | 0.4018         | 0.4077         | 0.2008                         | 0.2016         | 0.0661                | 0.0686         |
| 63                 | Attention | 0.8917                | 0.9819         | 0.4010         | 0.4064         | 0.1840                         | 0.1824         | 0.3067                | 0.3932         |

## Every recorded compiled kernel within each semantic stage

Calls are totals over the six retained rounds; times are per-round means. Compiler-fused operations remain indivisible. The tables do not invent times for separate constituents of a single GPU dispatch. Individual dispatch durations, layer IDs and round IDs are retained in the two evidence/final-study-\*.dispatches.json exports; they contain no token IDs, raw activations, absolute timestamps or process identifiers.

Expand the complete kernel breakdown

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Calls old / fixed | Old ms per round | Final fixed ms per round |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|--------------------------|
| drafter / Drafter<br>Cijk_Alik_Bljk_BBS_BH_Bias_HA_S_SAV_UserArgs_MT16x16x32_MI16x16x1_SN_LDSB1_AFC1_AG0_AGGSUA0_AGNTAB0_AFEM1_AFEM1_ASEM1_CD1_1_CLR0_CLS0_CADS0_DTLA0_DTLB0_DTLMO_DTVA0_DTVB1_DTVMSA0_DTVMSB0_DTVSM0_DPLB0_EPS0_ELFLR0_EMLLn1_FDSI0_GRPm1_GRVWA8_GRVWB8_GSUAMB_GLS0_HPLR0_ISA1201_ICIW0_IU1_K1_LDSTI0_LBSPPA128_LBSPPB0_LBSPPMXA0_LBSPPMXSBO_LBSPPM0_LPA16_LPB0_LPMXA0_LPMXSBO_LPM0_LRVW8_LWPMn1_MIAV1_MIWT1_1_MXLIBL_MXSFNs_MO40_MGRIPM1_NTN1_NTA0 NTB0_NTC0_NTD0_NTE0_NTMXA0_NTMXSBO_NTM0_NTWS0_NVn1_NVA0_NVBO_NVC0_NVD0_NVE0_NVMXA0_NVMXSBO_NVM0_NVWS0_NEPBS0_NLCA1_NLCB2_ONLL1_PAP0_PGL0_PGR1_PLR1_PKA0_SGROB0_SIA3_SS0_SPO0_SRVW0_SSO0_SVW8_SK0_SKFTR0_SKFDPO0_SKXCCM0_SNLL0_SIP1_SGRO0_TDMI0_TDMIM0_TDMS0_TIN0_Thn1_THA0_THB0_THC0_THD0_THE0_THMXSA0_THMXSBO_THM0_THWS0_TLDS1_TLDSM1_ULSGRO0_USL1_USLMX0_UIOFGRO0_UPLRP0_USFGR0n1_USI0_VSn1_VWA1_VWB1_WSGRA0_WSGRB0_WS32_WG16_2_1.kd   | 6 / 6             | 0.007179         | 0.007186                 |
| drafter / Drafter<br>Cijk_Alik_Bljk_BBS_BH_Bias_HA_S_SAV_UserArgs_MT16x32x256_MI16x16x1_SN_LDSB0_AFC1_AG0_AGGSUA0_AGNTAB0_AFEM1_AFEM1_ASEM1_CD1_1_CLR1_CLS0_CADS0_DTLA0_DTLB0_DTLMO_DTVA0_DTVB1_DTVMSA0_DTVMSB0_DTVSM0_DPLB0_EPS1_ELFLR0_EMLLn1_FDSI0_GRPm1_GRVWA8_GRVWB8_GSUAMB_GLS0_HPLR0_ISA1201_ICIW0_IU1_K1_LDSTI0_LBSPPA512_LBSPPB0_LBSPPMXA0_LBSPPMXSBO_LBSPPM0_LPA16_LPB0_LPMXA0_LPMXSBO_LPM0_LRVW8_LWPMn1_MIAV1_MIWT1_1_MXLIBL_MXSFNs_M040_MGRIPM1_NTN1_NTA0 NTB0_NTC0_NTD0_NTE0_NTMXA0_NTMXSBO_NTM0_NTWS0_NVn1_NVA0_NVBO_NVC0_NVD0_NVE0_NVMXA0_NVMXSBO_NVM0_NVWS0_NEPBS0_NLCA1_NLCB16_ONLL0_PAP0_PGL0_PGR1_PLR1_PKA0_SGROB0_SIA3_SS0_SPO0_SRVW0_SSO0_SVW8_SK0_SKFTR0_SKFDPO0_SKXCCM0_SNLL0_SIP1_SGRO0_TDMI0_TDMIM0_TDMS0_TIN0_Thn1_THA0_THB0_THC0_THD0_THE0_THMXSA0_THMXSBO_THM0_THWS0_TLDS1_TLDSM1_ULSGRO0_USL1_USLMX0_UIOFGRO0_UPLRP0_USFGR0n1_USI0_VSn1_VWA1_VWB1_WSGRA0_WSGRB0_WS32_WG16_4_1.kd | 6 / 6             | 0.177347         | 0.177167                 |
| drafter / Drafter<br>__amd_rocclr_copyBuffer.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6 / 6             | 0.001919         | 0.002012                 |
| drafter / Drafter<br>__amd_rocclr_fillBufferAligned.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 18 / 18           | 0.007077         | 0.006977                 |
| drafter / Drafter<br>_cache_draft_logits_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6 / 6             | 0.002359         | 0.002292                 |
| drafter / Drafter<br>_draft_head_int2.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6 / 6             | 0.824683         | 0.840683                 |
| drafter / Drafter<br>_gemm_a8w8_blockscale_preshuffle_kernel_GROUP_K_128_GROUP_N_128_BLOCK_SIZE_M_16_BLOCK_SIZE_N_128_BLOCK_SIZE_K_128_GROUP_SIZE_M_8_NUM_KSPLIT_1_SPLITK_BLOCK_SIZE_17408_EVEN_K_1_GRID_MN_40_cache_modifier_NONE.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 30 / 30           | 0.730700         | 0.739805                 |

| Scope / stage / compiled kernel                                                                                                                                                                                                       | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------|--------------------------------|
| drafter / Drafter<br>_gemm_a8w8_blockscale_preshuffle_kernel_GROUP_K_128_GROUP_N_128_BLOCK_SIZE_M_16_BLOCK_SIZE_N_128_BLOCK_SIZE_K_128_GROUP_SIZE_M_8_NUM_KSPLIT_1_SPLITK_BLOCK_SIZE_25600_EVEN_K_1_GRID_MN_40_cache_modifier_NONE.kd | 6 / 6                   | 0.218080            | 0.228567                       |
| drafter / Drafter<br>_gemm_a8w8_blockscale_preshuffle_kernel_GROUP_K_128_GROUP_N_128_BLOCK_SIZE_M_16_BLOCK_SIZE_N_128_BLOCK_SIZE_K_128_GROUP_SIZE_M_8_NUM_KSPLIT_1_SPLITK_BLOCK_SIZE_4096_EVEN_K_1_GRID_MN_40_cache_modifier_NONE.kd  | 30 /<br>30              | 0.202844            | 0.203789                       |
| drafter / Drafter<br>_gemm_a8w8_blockscale_preshuffle_kernel_GROUP_K_128_GROUP_N_128_BLOCK_SIZE_M_16_BLOCK_SIZE_N_128_BLOCK_SIZE_K_128_GROUP_SIZE_M_8_NUM_KSPLIT_1_SPLITK_BLOCK_SIZE_5120_EVEN_K_1_GRID_MN_272_cache_modifier_NONE.kd | 30 /<br>30              | 1.446181            | 1.447936                       |
| drafter / Drafter<br>_gemm_a8w8_blockscale_preshuffle_kernel_GROUP_K_128_GROUP_N_128_BLOCK_SIZE_M_16_BLOCK_SIZE_N_128_BLOCK_SIZE_K_128_GROUP_SIZE_M_8_NUM_KSPLIT_1_SPLITK_BLOCK_SIZE_5120_EVEN_K_1_GRID_MN_48_cache_modifier_NONE.kd  | 30 /<br>30              | 0.273222            | 0.274569                       |
| drafter / Drafter<br>_prepare_dflash_inputs_kernel.kd                                                                                                                                                                                 | 6 / 6                   | 0.007819            | 0.008032                       |
| drafter / Drafter<br>_rerank_exact.kd                                                                                                                                                                                                 | 6 / 6                   | 0.008606            | 0.008919                       |
| drafter / Drafter<br>_selector_walk_kernel.kd                                                                                                                                                                                         | 6 / 6                   | 0.007866            | 0.008092                       |
| drafter / Drafter<br>kernel_unified_attention.kd                                                                                                                                                                                      | 30 /<br>30              | 1.852924            | 1.957785                       |
| drafter / Drafter<br>reshape_and_cache_kernel_flash.kd                                                                                                                                                                                | 60 /<br>60              | 0.026984            | 0.027483                       |
| drafter / Drafter<br>triton_per_fused_4.kd                                                                                                                                                                                            | 6 / 6                   | 0.001926            | 0.001986                       |
| drafter / Drafter<br>triton_per_fused_8.kd                                                                                                                                                                                            | 0 / 24                  | 0.000000            | 0.007603                       |
| drafter / Drafter<br>triton_per_fused_9.kd                                                                                                                                                                                            | 24 / 0                  | 0.007490            | 0.000000                       |
| drafter / Drafter<br>triton_per_fused__to_copy_abs_clamp_div_max_preshuffle_gemm_squeeze_view_0.kd                                                                                                                                    | 30 /<br>30              | 0.009582            | 0.010142                       |
| drafter / Drafter<br>triton_per_fused__to_copy_abs_clamp_div_max_preshuffle_gemm_squeeze_view_2.kd                                                                                                                                    | 6 / 6                   | 0.002272            | 0.002086                       |
| drafter / Drafter<br>triton_per_fused__to_copy_abs_clamp_div_max_preshuffle_gemm_squeeze_view_4.kd                                                                                                                                    | 54 /<br>54              | 0.014478            | 0.016224                       |
| drafter / Drafter<br>triton_per_fused__to_copy_abs_clamp_div_max_view_0.kd                                                                                                                                                            | 6 / 6                   | 0.004099            | 0.005499                       |
| drafter / Drafter<br>triton_poi_fused_0.kd                                                                                                                                                                                            | 6 / 6                   | 0.002686            | 0.002606                       |
| drafter / Drafter<br>triton_poi_fused_10.kd                                                                                                                                                                                           | 24 / 0                  | 0.007503            | 0.000000                       |
| drafter / Drafter<br>triton_poi_fused_5.kd                                                                                                                                                                                            | 6 / 6                   | 0.002232            | 0.002319                       |
| drafter / Drafter<br>triton_poi_fused_9.kd                                                                                                                                                                                            | 0 / 24                  | 0.000000            | 0.009156                       |
| drafter / Drafter<br>triton_poi_fused__to_copy_clamp_div_mul_preshuffle_gemm_silu_slice_squeeze_view_7.kd                                                                                                                             | 30 / 0                  | 0.015428            | 0.000000                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------|--------------------------------|
| drafter / Drafter<br>triton_poi_fused__to_copy_clamp_div_preshuffle_gemm_squeeze_view_1.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 30 /<br>30              | 0.011795            | 0.011862                       |
| drafter / Drafter<br>triton_poi_fused__to_copy_clamp_div_preshuffle_gemm_squeeze_view_3.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6 / 6                   | 0.002059            | 0.002146                       |
| drafter / Drafter<br>triton_poi_fused__to_copy_clamp_div_preshuffle_gemm_squeeze_view_5.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 54 /<br>54              | 0.016371            | 0.016211                       |
| drafter / Drafter<br>triton_poi_fused_add_arange_bitwise_and_constant_pad_nd_fused_add_rms_norm_ge_mul_s<br>elect_slice_unsqueeze_view_3.kd                                                                                                                                                                                                                                                                                                                                                                                                                           | 54 /<br>54              | 0.019358            | 0.031305                       |
| drafter / Drafter<br>triton_poi_fused_add_arange_bitwise_and_constant_pad_nd_ge_mul_rms_norm_select_slic<br>e_unsqueeze_view_1.kd                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6 / 6                   | 0.002306            | 0.023693                       |
| drafter / Drafter<br>triton_poi_fused_add_permute_unsqueeze_view_2.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6 / 6                   | 0.001846            | 0.001846                       |
| drafter / Drafter<br>triton_poi_fused_cat_expand_index_mul_slice_unsqueeze_view_1.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6 / 6                   | 0.002546            | 0.002539                       |
| drafter / Drafter<br>triton_red_fused__to_copy_abs_clamp_div_max_mul_preshuffle_gemm_silu_slice_squeeze_<br>view_6.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30 /<br>30              | 0.013422            | 0.018855                       |
| drafter / Drafter<br>triton_red_fused__to_copy_add_arange_bitwise_and_constant_pad_nd_fused_add_rms_norm<br>_ge_mul_select_slice_unsqueeze_view_w4_gemm_2.kd                                                                                                                                                                                                                                                                                                                                                                                                          | 30 /<br>30              | 0.030876            | 0.032549                       |
| drafter / Drafter<br>triton_red_fused__to_copy_add_arange_bitwise_and_constant_pad_nd_fused_add_rms_norm<br>_ge_mul_select_slice_unsqueeze_view_w4_gemm_7.kd                                                                                                                                                                                                                                                                                                                                                                                                          | 0 / 24                  | 0.000000            | 0.027043                       |
| drafter / Drafter<br>triton_red_fused__to_copy_add_arange_bitwise_and_constant_pad_nd_fused_add_rms_norm<br>_ge_mul_select_slice_unsqueeze_view_w4_gemm_8.kd                                                                                                                                                                                                                                                                                                                                                                                                          | 24 / 0                  | 0.025756            | 0.000000                       |
| drafter / Drafter<br>triton_red_fused__to_copy_embedding_mul_rms_norm_w4_gemm_0.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6 / 6                   | 0.003846            | 0.004519                       |
| drafter / Drafter<br>triton_red_fused_add_arange_bitwise_and_constant_pad_nd_fused_add_rms_norm_ge_mul_s<br>elect_slice_unsqueeze_view_7.kd                                                                                                                                                                                                                                                                                                                                                                                                                           | 0 / 6                   | 0.000000            | 0.006599                       |
| drafter / Drafter<br>triton_red_fused_add_arange_bitwise_and_constant_pad_nd_fused_add_rms_norm_ge_mul_s<br>elect_slice_unsqueeze_view_8.kd                                                                                                                                                                                                                                                                                                                                                                                                                           | 6 / 0                   | 0.006719            | 0.000000                       |
| drafter / Drafter<br>void at::native::(anonymous namespace)::CatArrayBatchedCopy_contig<at::native::<br>(anonymous namespace)::OpaqueType<2u>, unsigned int, 2, 128, 1>(at::native::<br>(anonymous namespace)::OpaqueType<2u>*, at::native::(anonymous<br>namespace)::CatArrInputTensorMetadata<at::native::(anonymous<br>namespace)::OpaqueType<2u>, unsigned int, 128, 1>, at::native::(anonymous<br>namespace)::TensorSizeStride<unsigned int, 4u>, int, unsigned int) [clone .kd]                                                                                 | 12 /<br>12              | 0.009285            | 0.009498                       |
| drafter / Drafter<br>void at::native::_scatter_gather_elementwise_kernel<256, 4,<br>at::native::_cuda_scatter_gather_internal_kernel<false, at::native::OpaqueType<4>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}>(int,<br>at::native::_cuda_scatter_gather_internal_kernel<false, at::native::OpaqueType<4>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}) [clone .kd] | 6 / 6                   | 0.003526            | 0.003532                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------|--------------------------------|
| drafter / Drafter<br>void at::native::_scatter_gather_elementwise_kernel<256, 4,<br>at::native::_cuda_scatter_gather_internal_kernel<true, at::native::OpaqueType<2>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}>(int,<br>at::native::_cuda_scatter_gather_internal_kernel<true, at::native::OpaqueType<2>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 / 6                   | 0.002879            | 0.003066                       |
| drafter / Drafter<br>void at::native::bitonicSortKVInPlace<2, -1, 16, 16, c10::BFloat16, long,<br>at::native::GTOp<c10::BFloat16, true>, unsigned<br>int>(at::cuda::detail::TensorInfo<c10::BFloat16, unsigned int>, unsigned int,<br>unsigned int, unsigned int, at::cuda::detail::TensorInfo<long, unsigned int>,<br>unsigned int, at::native::GTOp<c10::BFloat16, true>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6 / 6                   | 0.003232            | 0.003306                       |
| drafter / Drafter<br>void at::native::elementwise_kernel_manual_unroll<128, 4,<br>at::native::gpu_kernel_impl<at::native::direct_copy_kernel_cuda(at::TensorIterat<br>orBase&)::{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1} const&)::{lambda(int, bool)#1}>(int,<br>at::native::gpu_kernel_impl<at::native::direct_copy_kernel_cuda(at::TensorIterat<br>orBase&)::{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1} const&)::{lambda(int, bool)#1}) [clone .kd]                                                       | 6 / 6                   | 0.005012            | 0.005039                       |
| drafter / Drafter<br>void at::native::elementwise_kernel_manual_unroll<128, 4,<br>at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIt<br>eratorBase&)::{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1} const&)::{lambda(int, bool)#1}>(int,<br>at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIt<br>eratorBase&)::{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()#3}::operator()() const::{lambda()#4}::operator()() const::<br>{lambda(long)#1} const&)::{lambda(int, bool)#1}) [clone .kd]                                         | 6 / 6                   | 0.002586            | 0.002546                       |
| drafter / Drafter<br>void at::native::elementwise_kernel_manual_unroll<128, 8,<br>at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIt<br>eratorBase&)::{lambda()#3}::operator()() const::{lambda()#12}::operator()() const::<br>{lambda(c10::BFloat16)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()#3}::operator()() const::{lambda()#12}::operator()() const::<br>{lambda(c10::BFloat16)#1} const&)::{lambda(int, bool)#1}>(int,<br>at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIt<br>eratorBase&)::{lambda()#3}::operator()() const::{lambda()#12}::operator()() const::<br>{lambda(c10::BFloat16)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()#3}::operator()() const::{lambda()#12}::operator()() const::<br>{lambda(c10::BFloat16)#1} const&)::{lambda(int, bool)#1}) [clone .kd] | 6 / 6                   | 0.003506            | 0.003692                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| drafter / Drafter<br>void at::native::index_elementwise_kernel<128, 4,<br>at::native::gpu_index_kernel<at::native::index_kernel_impl<at::native::OpaqueType<4<br>>>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>{lambda(char*, char const*, long)#1}>(at::TensorIteratorBase&, c10::ArrayRef<long>,<br>c10::ArrayRef<long>, at::native::index_kernel_impl<at::native::OpaqueType<4><br>>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>{lambda(char*, char const*, long)#1} const&, bool)::{lambda(int)#1}>(long,<br>at::native::gpu_index_kernel<at::native::index_kernel_impl<at::native::OpaqueType<4<br>>>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>{lambda(char*, char const*, long)#1}>(at::TensorIteratorBase&, c10::ArrayRef<long>,<br>c10::ArrayRef<long>, at::native::index_kernel_impl<at::native::OpaqueType<4><br>>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>{lambda(char*, char const*, long)#1} const&, bool)::{lambda(int)#1}) [clone .kd] | 6 / 6                   | 0.003146            | 0.003146                       |
| drafter / Drafter<br>void at::native::mbtopk::computeBlockDigitCounts<c10::BFloat16, unsigned int,<br>unsigned int, 2>(at::cuda::detail::TensorInfo<c10::BFloat16 const, unsigned int>,<br>unsigned int, unsigned int, unsigned int, unsigned int, int, int, unsigned int,<br>unsigned int, unsigned int*, short*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 12 /<br>12              | 0.038345            | 0.041232                       |
| drafter / Drafter<br>void at::native::mbtopk::computeBlockDigitCounts<float, unsigned int, unsigned int,<br>2>(at::cuda::detail::TensorInfo<float const, unsigned int>, unsigned int, unsigned<br>int*, unsigned int, unsigned int, int, int, unsigned int, unsigned int, unsigned<br>int*, short*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 24 /<br>24              | 0.055463            | 0.057470                       |
| drafter / Drafter<br>void at::native::mbtopk::computeBlockwiseWithinKCounts<unsigned int,<br>c10::BFloat16>(unsigned int*, short*, unsigned int*, unsigned int, int, bool,<br>unsigned int*, c10::BFloat16*, unsigned int*, unsigned int*, unsigned int*,<br>unsigned int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 12 /<br>12              | 0.022351            | 0.022578                       |
| drafter / Drafter<br>void at::native::mbtopk::computeBlockwiseWithinKCounts<unsigned int,<br>float>(unsigned int*, short*, unsigned int*, unsigned int, int, bool, unsigned<br>int*, float*, unsigned int*, unsigned int*, unsigned int*, unsigned int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24 /<br>24              | 0.028736            | 0.029063                       |
| drafter / Drafter<br>void at::native::mbtopk::fill<unsigned int, unsigned int>(unsigned int*, unsigned<br>int, unsigned int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12 /<br>12              | 0.003018            | 0.003005                       |
| drafter / Drafter<br>void at::native::mbtopk::gatherTopK<c10::BFloat16, unsigned int,<br>2>(at::cuda::detail::TensorInfo<c10::BFloat16 const, unsigned int>, unsigned int,<br>unsigned int, bool, unsigned int, unsigned int,<br>at::cuda::detail::TensorInfo<c10::BFloat16, unsigned int>, unsigned int,<br>at::cuda::detail::TensorInfo<long, unsigned int>, unsigned int, unsigned int,<br>unsigned int, c10::BFloat16*, unsigned int*, unsigned int*, unsigned int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6 / 6                   | 0.021039            | 0.021259                       |
| drafter / Drafter<br>void at::native::mbtopk::gatherTopK<float, unsigned int,<br>2>(at::cuda::detail::TensorInfo<float const, unsigned int>, unsigned int, unsigned<br>int, bool, unsigned int, unsigned int, at::cuda::detail::TensorInfo<float, unsigned<br>int>, unsigned int, at::cuda::detail::TensorInfo<long, unsigned int>, unsigned int,<br>unsigned int, unsigned int, float*, unsigned int*, unsigned int*, unsigned int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6 / 6                   | 0.008879            | 0.013026                       |
| drafter / Drafter<br>void at::native::reduce_kernel<512, 1, at::native::ReduceOp<float,<br>at::native::func_wrapper_t<float, at::native::sum_functor<float, float,<br>float>::operator()(at::TensorIterator&)::{lambda(float, float)#1}>, unsigned int,<br>float, 4, 4>>(at::native::ReduceOp<float, at::native::func_wrapper_t<float,<br>at::native::sum_functor<float, float, float>::operator()(at::TensorIterator&)::<br>{lambda(float, float)#1}>, unsigned int, float, 4, 4>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6 / 6                   | 0.003759            | 0.003799                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                       | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| drafter / Drafter<br>void at::native::vectorized_elementwise_kernel<4,<br>at::native::CUDAFunctionOnSelf_add<long>, std::array<char*, 2ul> >(int,<br>at::native::CUDAFunctionOnSelf_add<long>, std::array<char*, 2ul>) [clone .kd]                                                                                                                                    | 6 / 6                   | 0.001732            | 0.001866                       |
| drafter / Drafter<br>void at::native::vectorized_elementwise_kernel<4,<br>at::native::bfloat16_copy_kernel_cuda(at::TensorIteratorBase&)::{lambda(float)#1},<br>std::array<char*, 2ul> >(int,<br>at::native::bfloat16_copy_kernel_cuda(at::TensorIteratorBase&)::{lambda(float)#1},<br>std::array<char*, 2ul>) [clone .kd]                                            | 6 / 6                   | 0.002479            | 0.002486                       |
| drafter / Drafter<br>void at::native::vectorized_elementwise_kernel<4,<br>at::native::bfloat16tofloat32_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda(c10::BFloat16)#1}, std::array<char*, 2ul> >(int,<br>at::native::bfloat16tofloat32_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda(c10::BFloat16)#1}, std::array<char*, 2ul>) [clone .kd]        | 12 /<br>12              | 0.003785            | 0.003858                       |
| drafter / Drafter<br>void at::native::vectorized_elementwise_kernel<8,<br>at::native::FillFunction<c10::BFloat16>, std::array<char*, 1ul> >(int,<br>at::native::FillFunction<c10::BFloat16>, std::array<char*, 1ul>) [clone .kd]                                                                                                                                      | 12 /<br>12              | 0.005351            | 0.005218                       |
| drafter / Drafter<br>void at::native::vectorized_gather_kernel<16, long>(char*, char*, long*, int, long,<br>long, long, long, bool) [clone .kd]                                                                                                                                                                                                                       | 6 / 6                   | 0.002099            | 0.002139                       |
| drafter / Drafter<br>void at::native::warpMergeSortKVInPlace<2, -1, 128, 16, float, long,<br>at::native::GT0p<float, true>, unsigned int,<br>32>(at::cuda::detail::TensorInfo<float, unsigned int>, unsigned int, unsigned int,<br>unsigned int, at::cuda::detail::TensorInfo<long, unsigned int>, unsigned int,<br>at::native::GT0p<float, true>, float) [clone .kd] | 6 / 6                   | 0.004952            | 0.005019                       |
| drafter / Drafter<br>void r4d_gemm_w4a16_nt_m64_kernel<1, 1, false>(unsigned short const*, unsigned int<br>const*, unsigned int const*, __hip_bfloat16*, int, int, int, int, int) [clone .kd]                                                                                                                                                                         | 6 / 6                   | 0.004506            | 0.004599                       |
| drafter / Drafter<br>void r4d_gemm_w4a16_nt_m64_kernel<1, 1, true>(unsigned short const*, unsigned int<br>const*, unsigned int const*, __hip_bfloat16*, int, int, int, int, int) [clone .kd]                                                                                                                                                                          | 60 /<br>60              | 0.080050            | 0.081410                       |
| drafter / Drafter<br>void vllm::rms_norm_kernel<c10::BFloat16, 8, 2, true>(c10::BFloat16*, c10::BFloat16<br>const*, long, long, long, long, long, c10::BFloat16 const*, long, float, int, int)<br>[clone .kd]                                                                                                                                                         | 6 / 6                   | 0.003059            | 0.003192                       |
| drafter / Drafter<br>void vllm::rms_norm_kernel<c10::BFloat16, 8, 4, true>(c10::BFloat16*, c10::BFloat16<br>const*, long, long, long, long, long, c10::BFloat16 const*, long, float, int, int)<br>[clone .kd]                                                                                                                                                         | 6 / 6                   | 0.002752            | 0.002779                       |
| drafter / Drafter<br>void vllm::rotary_embedding_kernel<c10::BFloat16, c10::BFloat16, true>(long const*,<br>c10::BFloat16*, c10::BFloat16*, c10::BFloat16 const*, int, long, long, long, int,<br>int, int, long, bool) [clone .kd]                                                                                                                                    | 6 / 6                   | 0.002499            | 0.002579                       |
| target_body / Attention KV write<br>reshape_and_cache_kernel_flash.kd                                                                                                                                                                                                                                                                                                 | 96 /<br>96              | 0.046324            | 0.046878                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>triton_poi_fused_1.kd                                                                                                                                                                                                                                                                                   | 0 / 96                  | 0.000000            | 0.024404                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>triton_poi_fused_3.kd                                                                                                                                                                                                                                                                                   | 0 / 96                  | 0.000000            | 0.027984                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>triton_poi_fused_4.kd                                                                                                                                                                                                                                                                                   | 0 / 96                  | 0.000000            | 0.028471                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>triton_poi_fused_6.kd                                                                                                                                                                                                                                                                                   | 96 / 0                  | 0.022437            | 0.000000                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| target_body / Attention Q/K normalization, RoPE and layout<br>triton_poi_fused_8.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 96 / 0                  | 0.030144            | 0.000000                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>triton_poi_fused_arange_bitwise_and_eq_index_lt_remainder_select_split_where_2.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0 / 96                  | 0.000000            | 0.029938                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>triton_red_fused_7.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 96 / 0                  | 0.026091            | 0.000000                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>void at::native::elementwise_kernel_manual_unroll<128, 8,<br>at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIter<br>atorBase&)::{lambda()}#3>::operator>()() const:: {lambda()}#12>::operator>()() const::<br>{lambda(c10::BFloat16)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()}#3>::operator>()() const:: {lambda()}#12>::operator>()() const::<br>{lambda(c10::BFloat16)#1} const&:: {lambda(int, bool)#1}>(int,<br>at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIt<br>eratorBase&)::{lambda()}#3>::operator>()() const:: {lambda()}#12>::operator>()() const::<br>{lambda(c10::BFloat16)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>{lambda()}#3>::operator>()() const:: {lambda()}#12>::operator>()() const::<br>{lambda(c10::BFloat16)#1} const&:: {lambda(int, bool)#1}) [clone .kd] | 0 / 96                  | 0.000000            | 0.033671                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>void stock_m1_gemma_norm<false, 256, 32>(unsigned short const*, unsigned short<br>const*, unsigned short const*, unsigned short*, unsigned short*, long, long, float,<br>float*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0 / 96                  | 0.000000            | 0.050797                       |
| target_body / Attention Q/K normalization, RoPE and layout<br>void stock_m1_gemma_norm<false, 256, 64>(unsigned short const*, unsigned short<br>const*, unsigned short const*, unsigned short*, unsigned short*, long, long, float,<br>float*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0 / 96                  | 0.000000            | 0.040311                       |
| target_body / Attention decode<br>void qwen_stock_m1_shared_decode<4, 16, 256, 6, 16, 0, 3430971>(R4DArgs, int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0 / 96                  | 0.000000            | 5.285290                       |
| target_body / Attention decode<br>void r4d_attn_decode_kernel<3, 16, 256, 6, 16, 0, 3430971>(R4DArgs, int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 96 / 0                  | 4.224365            | 0.000000                       |
| target_body / Attention input activation FP8 quantization<br>void vllm::dynamic_per_token_scaled_fp8_quant_kernel_strided<c10::BFloat16,<br>c10::Float8_e4m3fn>(c10::Float8_e4m3fn*, float*, c10::BFloat16 const*, float<br>const*, int, long, long) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 96 /<br>96              | 0.042844            | 0.042158                       |
| target_body / Attention input projection<br>void radiancexfp4_fp8_gemm_decode<8, 128, 1, 1, true, true, true>(unsigned char<br>const*, unsigned char const*, unsigned char const*, unsigned char const*, float<br>const*, float*, int*, std::bfloat16_t*, int, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 96 /<br>96              | 1.099262            | 1.098923                       |
| target_body / Attention output activation FP8 quantization<br>void vllm::dynamic_per_token_scaled_fp8_quant_kernel_strided<c10::BFloat16,<br>c10::Float8_e4m3fn>(c10::Float8_e4m3fn*, float*, c10::BFloat16 const*, float<br>const*, int, long, long) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 96 /<br>96              | 0.045851            | 0.045791                       |
| target_body / Attention output gating<br>triton_poi_fused_mul_mxfp4_linear_sigmoid_view_0.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 96 /<br>96              | 0.028144            | 0.032364                       |
| target_body / Attention output projection<br>void radiancexfp4_fp8_gemm_decode<8, 128, 4, 1, true, true, true>(unsigned char<br>const*, unsigned char const*, unsigned char const*, unsigned char const*, float<br>const*, float*, int*, std::bfloat16_t*, int, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 96 /<br>96              | 0.556920            | 0.521433                       |
| target_body / Attention split-KV merge<br>void qwen_stock_m1_shared_merge<256, 4, 1>(R4DArgs, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0 / 96                  | 0.000000            | 0.119631                       |
| target_body / Attention split-KV merge<br>void r4d_attn_splitkv_combine_kernel<256, 4, 1>(R4DArgs, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 96 / 0                  | 0.137671            | 0.000000                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| target_body / Embedding + first input normalization<br>triton_poi_fused__to_copy_embedding_0.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0 / 6                   | 0.000000            | 0.002599                       |
| target_body / Embedding + first input normalization<br>triton_red_fused__to_copy_add_embedding_mxfp4_linear_rms_norm_0.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 / 0                   | 0.004112            | 0.000000                       |
| target_body / Embedding + first input normalization<br>void stock_m1_gemma_norm<false, 5120, 512>(unsigned short const*, unsigned short const*, unsigned short const*, unsigned short*, unsigned short*, long, long, float, float*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0 / 6                   | 0.000000            | 0.005159                       |
| target_body / Final normalization/layout<br>triton_red_fused__to_copy_add_fused_add_rms_norm_3.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6 / 0                   | 0.002392            | 0.000000                       |
| target_body / Final normalization/layout<br>void stock_m1_gemma_norm<true, 5120, 512>(unsigned short const*, unsigned short const*, unsigned short const*, unsigned short*, unsigned short*, long, long, float, float*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0 / 6                   | 0.000000            | 0.005612                       |
| target_body / GDN convolution<br>_causal_conv1d_update_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0 /<br>288              | 0.000000            | 0.226365                       |
| target_body / GDN convolution<br>void r4d_gdn_conv_update_kernel<1>(unsigned short const*, long, unsigned short const*, unsigned short const*, unsigned short*, long, long, long, int, int const*, long, int const*, unsigned short*, unsigned short*, unsigned short*, int const*, int, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 288 /<br>0              | 0.494015            | 0.000000                       |
| target_body / GDN input activation FP8 quantization<br>void vllm::dynamic_per_token_scaled_fp8_quant_kernel_strided<c10::BFloat16, c10::Float8_e4m3fn>(c10::Float8_e4m3fn*, float*, c10::BFloat16 const*, float const*, int, long, long) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 288 /<br>288            | 0.127500            | 0.127346                       |
| target_body / GDN input projection<br>void radiance_mxfp4_fp8_gemm_decode<8, 128, 1, 1, true, true, true>(unsigned char const*, unsigned char const*, unsigned char const*, unsigned char const*, float const*, float*, int*, std::bfloat16_t*, int, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 288 /<br>288            | 3.875011            | 3.872797                       |
| target_body / GDN layout/copies and buffer initialization<br>triton_per_fused_1.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 96 / 0                  | 0.022318            | 0.000000                       |
| target_body / GDN layout/copies and buffer initialization<br>triton_poi_fused_0.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 96 / 0                  | 0.031257            | 0.000000                       |
| target_body / GDN layout/copies and buffer initialization<br>triton_poi_fused_add_1.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 / 18                  | 0.000000            | 0.004004                       |
| target_body / GDN layout/copies and buffer initialization<br>triton_poi_fused_add_2.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 / 12                  | 0.000000            | 0.002738                       |
| target_body / GDN layout/copies and buffer initialization<br>void at::native::elementwise_kernel_manual_unroll<128, 8, at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::{lambda()#3}::operator()() const:: {lambda()#12}::operator()() const:: {lambda(c10::BFloat16)#1}>(at::TensorIteratorBase&, at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&):: {lambda()#3}::operator()() const:: {lambda()#12}::operator()() const:: {lambda(c10::BFloat16)#1} const&)::{lambda(int, bool)#1}>(int, at::native::gpu_kernel_impl_nocast<at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::{lambda()#3}::operator()() const:: {lambda()#12}::operator()() const:: {lambda(c10::BFloat16)#1}>(at::TensorIteratorBase&, at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&):: {lambda()#3}::operator()() const:: {lambda()#12}::operator()() const:: {lambda(c10::BFloat16)#1} const&)::{lambda(int, bool)#1}) [clone .kd] | 288 /<br>288            | 0.156100            | 0.157119                       |
| target_body / GDN layout/copies and buffer initialization<br>void at::native::vectorized_elementwise_kernel<4, at::native::bfloat16_copy_kernel_cuda(at::TensorIteratorBase&)::{lambda(float)#1}, std::array<char*, 2ul> >(int, at::native::bfloat16_copy_kernel_cuda(at::TensorIteratorBase&)::{lambda(float)#1}, std::array<char*, 2ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 /<br>288              | 0.000000            | 0.092759                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                          | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| target_body / GDN layout/copies and buffer initialization<br>void at::native::vectorized_elementwise_kernel<8,<br>at::native::FillFunctor<c10::BFloat16>, std::array<char*, 1ul> >(int,<br>at::native::FillFunctor<c10::BFloat16>, std::array<char*, 1ul>) [clone .kd]                                                                                                                                   | 288 /<br>288            | 0.074632            | 0.074886                       |
| target_body / GDN output activation FP8 quantization<br>void vllm::dynamic_per_token_scaled_fp8_quant_kernel_strided<c10::BFloat16,<br>c10::Float8_e4m3fn>(c10::Float8_e4m3fn*, float*, c10::BFloat16 const*, float<br>const*, int, long, long) [clone .kd]                                                                                                                                              | 288 /<br>288            | 0.128079            | 0.131086                       |
| target_body / GDN output gated normalization<br>layer_norm_fwd_kernel.kd                                                                                                                                                                                                                                                                                                                                 | 0 /<br>288              | 0.000000            | 0.097307                       |
| target_body / GDN output gated normalization<br>triton_per_fused__to_copy_mean_pow_view_0.kd                                                                                                                                                                                                                                                                                                             | 192 /<br>0              | 0.050375            | 0.000000                       |
| target_body / GDN output gated normalization<br>triton_poi_fused__to_copy_add_mean_mul_mxfp4_linear_pow_rsqrtsilu_view_1.kd                                                                                                                                                                                                                                                                              | 192 /<br>0              | 0.058742            | 0.000000                       |
| target_body / GDN output gated normalization<br>triton_poi_fused__to_copy_add_mean_mul_mxfp4_linear_pow_rsqrtsilu_view_2.kd                                                                                                                                                                                                                                                                              | 96 / 0                  | 0.028797            | 0.000000                       |
| target_body / GDN output projection<br>void radiance_mxfp4_fp8_gemm_decode<8, 128, 4, 1, true, true, true>(unsigned char<br>const*, unsigned char const*, unsigned char const*, unsigned char const*, float<br>const*, float*, int*, std::bfloat16_t*, int, int, int) [clone .kd]                                                                                                                        | 288 /<br>288            | 1.884129            | 1.883793                       |
| target_body / GDN recurrence and gates<br>stock_gdn_scan_kernel.kd                                                                                                                                                                                                                                                                                                                                       | 0 /<br>288              | 0.000000            | 1.218805                       |
| target_body / GDN recurrence and gates<br>void r4d_gdn_recurrent_update_kernel<1, 0, 2>(unsigned short const*, unsigned short<br>const*, unsigned short const*, void const*, void const*, long, float const*, float<br>const*, float*, long, long, unsigned short*, int const*, int const*, long, int<br>const*, unsigned short const*, float const*, float, int, int, int, float, float)<br>[clone .kd] | 288 /<br>0              | 1.161877            | 0.000000                       |
| target_body / Layer input residual/normalization<br>triton_red_fused__to_copy_add_fused_add_rms_norm_mxfp4_linear_3.kd                                                                                                                                                                                                                                                                                   | 90 / 0                  | 0.037245            | 0.000000                       |
| target_body / Layer input residual/normalization<br>triton_red_fused__to_copy_add_fused_add_rms_norm_mxfp4_linear_4.kd                                                                                                                                                                                                                                                                                   | 192 /<br>0              | 0.078508            | 0.000000                       |
| target_body / Layer input residual/normalization<br>triton_red_fused__to_copy_add_fused_add_rms_norm_mxfp4_linear_5.kd                                                                                                                                                                                                                                                                                   | 96 / 0                  | 0.039184            | 0.000000                       |
| target_body / Layer input residual/normalization<br>void stock_m1_gemma_norm<true, 5120, 512>(unsigned short const*, unsigned short<br>const*, unsigned short const*, unsigned short*, unsigned short*, long, long, float,<br>float*) [clone .kd]                                                                                                                                                        | 0 /<br>378              | 0.000000            | 0.345972                       |
| target_body / MLP SiLU and gating<br>triton_poi_fused_mul_mxfp4_linear_silu_slice_0.kd                                                                                                                                                                                                                                                                                                                   | 0 /<br>288              | 0.000000            | 0.141272                       |
| target_body / MLP SiLU and gating<br>triton_poi_fused_mul_mxfp4_linear_silu_slice_1.kd                                                                                                                                                                                                                                                                                                                   | 0 / 96                  | 0.000000            | 0.048204                       |
| target_body / MLP SiLU and gating<br>triton_poi_fused_mul_mxfp4_linear_silu_slice_2.kd                                                                                                                                                                                                                                                                                                                   | 96 / 0                  | 0.044918            | 0.000000                       |
| target_body / MLP SiLU and gating<br>triton_poi_fused_mul_mxfp4_linear_silu_slice_3.kd                                                                                                                                                                                                                                                                                                                   | 192 /<br>0              | 0.066262            | 0.000000                       |
| target_body / MLP SiLU and gating<br>triton_poi_fused_mul_mxfp4_linear_silu_slice_4.kd                                                                                                                                                                                                                                                                                                                   | 96 / 0                  | 0.045138            | 0.000000                       |
| target_body / MLP down input FP8 quantization<br>void vllm::dynamic_per_token_scaled_fp8_quant_kernel_strided<c10::BFloat16,<br>c10::Float8_e4m3fn>(c10::Float8_e4m3fn*, float*, c10::BFloat16 const*, float<br>const*, int, long, long) [clone .kd]                                                                                                                                                     | 384 /<br>384            | 0.296558            | 0.297344                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| target_body / MLP down projection<br>void radiance_mxfp4_fp8_gemm_decode<8, 128, 4, 1, true, true, true>(unsigned char const*, unsigned char const*, unsigned char const*, unsigned char const*, float const*, float*, int*, std::bfloat16_t*, int, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 384 /<br>384            | 5.147348            | 5.142933                       |
| target_body / MLP gate/up input FP8 quantization<br>void vllm::dynamic_per_token_scaled_fp8_quant_kernel_strided<c10::BFloat16, c10::Float8_e4m3fn>(c10::Float8_e4m3fn*, float*, c10::BFloat16 const*, float const*, int, long, long) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 384 /<br>384            | 0.164856            | 0.166543                       |
| target_body / MLP gate/up projection<br>void radiance_mxfp4_fp8_gemm_folded<2, true, true>(unsigned char const*, unsigned char const*, unsigned char const*, unsigned char const*, float const*, std::bfloat16_t*, int, int, int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 384 /<br>384            | 25.271205           | 25.437279                      |
| target_body / Post-attention/GDN residual/normalization<br>triton_red_fused__to_copy_add_fused_add_rms_norm_mxfp4_linear_1.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 96 / 0                  | 0.035164            | 0.000000                       |
| target_body / Post-attention/GDN residual/normalization<br>triton_red_fused__to_copy_add_fused_add_rms_norm_mxfp4_linear_2.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 192 /<br>0              | 0.064875            | 0.000000                       |
| target_body / Post-attention/GDN residual/normalization<br>triton_red_fused__to_copy_add_fused_add_rms_norm_mxfp4_linear_3.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 96 / 0                  | 0.042258            | 0.000000                       |
| target_body / Post-attention/GDN residual/normalization<br>void stock_m1_gemma_norm<true, 5120, 512>(unsigned short const*, unsigned short const*, unsigned short const*, unsigned short*, unsigned short*, long, long, float, float*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0 /<br>384              | 0.000000            | 0.352599                       |
| target_vocabulary_head / Full BF16 target head<br>(anonymous namespace)::stock_m1_head_pair(__hip_bfloat16 const*, __hip_bfloat16 const*, __hip_bfloat16*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0 / 6                   | 0.000000            | 4.139846                       |
| target_vocabulary_head / Full BF16 target head<br>Cijk_Alik_Bljk_BBS_BH_Bias_HA_S_SAV_UserArgs_MT16x32x256_MI16x16x1_SN_LDSB0_AFC1_AG0_AGGSUA0_AGNTAB0_AFEM1_AFEM1_ASEM1_CD1_1_CLR1_CLS0_CADS0_DTLA0_DTLB0_DTLMO_DTVAA0_DTVB1_DTVMSA0_DTVMSB0_DTVSM0_DPLB0_EPS1_ELFLR0_EMLLn1_FDSI0_GRP1_GRVWA8_GRVWB8_GS_UAMB_GLS0_HPLR0_ISA1201_ICIW0_IU1_K1_LDSTI0_LBSPPA512_LBSPPB0_LBSPPMXSA0_LBSPPMXSB0_LBSPPM0_LPA16_LPB0_LPMXSA0_LPMXSB0_LPM0_LRVW8_LWPMn1_MIAV1_MIWT1_1_MXLIBL_MXSFNS_M040_MGRIPM1_NTN1_NTA0 NTB0_NTC0_NTD0_NTE0_NTMXSA0_NTMXSB0_NTM0_NTWS0_NVn1_NVA0_NVB0_NVC0_NVD0_NVE0_NVMXSA0_NVMXSB0_NVM0_NVWS0_NEPBS0_NLCA1_NLCB16_ONLL0_PAP0_PGL0_PGR1_PLR1_PKA0_SGROB0_SIA3_SS1_SPO0_SRVW0_SSO0_SVW1_SK0_SKFTR0_SKFDP00_SKXCCM0_SNLL0_SI_P1_SGRO0_TDMI0_TDMIM0_TDMS0_TIN0_THn1_THA0_THB0_THC0_THD0_THE0_THMXSA0_THMXSB0_THM0_THWS0_TLDS1_TLDSM1_ULSGRO0_USL1_USLMX0_UIOFGRO0_UPLRP0_USFGR0n1_USI0_VSn1_VWA1_VWB1_WSGRA0_WSGRB0_WS32_WG16_4_1.kd | 6 / 0                   | 4.024325            | 0.000000                       |
| unattributed / Other GPU bookkeeping<br>__amd_rocclr_copyBuffer.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 172 /<br>172            | 0.075278            | 0.076165                       |
| unattributed / Other GPU bookkeeping<br>__amd_rocclr_fillBufferAligned.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6 / 6                   | 0.002699            | 0.002519                       |
| unattributed / Other GPU bookkeeping<br>_combine_sampled_and_draft_tokens_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 7 / 7                   | 0.003432            | 0.003485                       |
| unattributed / Other GPU bookkeeping<br>_compute_local_logits_stats_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6 / 6                   | 0.028659            | 0.028819                       |
| unattributed / Other GPU bookkeeping<br>_compute_slot_mappings_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 7 / 7                   | 0.003386            | 0.003485                       |
| unattributed / Other GPU bookkeeping<br>_expand_idx_mapping_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 7 / 7                   | 0.002185            | 0.002339                       |
| unattributed / Other GPU bookkeeping<br>_gather_block_tables_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 7 / 7                   | 0.005632            | 0.005852                       |
| unattributed / Other GPU bookkeeping<br>_get_num_sampled_and_rejected_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6 / 6                   | 0.002632            | 0.002792                       |
| unattributed / Other GPU bookkeeping<br>_insert_resampled_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6 / 6                   | 0.003466            | 0.003499                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| unattributed / Other GPU bookkeeping<br>_post_update_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 6 / 6                   | 0.005292            | 0.005206                       |
| unattributed / Other GPU bookkeeping<br>_prepare_pos_seq_lens_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7 / 7                   | 0.002499            | 0.002392                       |
| unattributed / Other GPU bookkeeping<br>_prepare_rope_positions_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 7 / 7                   | 0.003119            | 0.003232                       |
| unattributed / Other GPU bookkeeping<br>_rejection_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 6 / 6                   | 0.007559            | 0.007219                       |
| unattributed / Other GPU bookkeeping<br>_resample_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6 / 6                   | 0.014159            | 0.015366                       |
| unattributed / Other GPU bookkeeping<br>_scatter_num_accepted_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6 / 6                   | 0.001939            | 0.001986                       |
| unattributed / Other GPU bookkeeping<br>postprocess_mamba_fused_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6 / 6                   | 0.002552            | 0.002699                       |
| unattributed / Other GPU bookkeeping<br>precopy_mamba_align_fused_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 7 / 7                   | 0.003079            | 0.003099                       |
| unattributed / Other GPU bookkeeping<br>preprocess_mamba_align_fused_kernel.kd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 7 / 7                   | 0.003119            | 0.003012                       |
| unattributed / Other GPU bookkeeping<br>void (anonymous namespace)::elementwise_kernel_with_index<int,<br>at::native::arange_cuda_out(c10::Scalar const&, c10::Scalar const&, c10::Scalar<br>const&, at::Tensor&)::{lambda()#1}::operator()() const:: {lambda()#3}::operator()()<br>const:: {lambda(long)#1}>(int, at::native::arange_cuda_out(c10::Scalar const&,<br>c10::Scalar const&, c10::Scalar const&, at::Tensor&)::{lambda()#1}::operator()()<br>const:: {lambda()#3}::operator()() const:: {lambda(long)#1},<br>function_traits<at::native::arange_cuda_out(c10::Scalar const&, c10::Scalar const&,<br>c10::Scalar const&, at::Tensor&)::{lambda()#1}::operator()() const::<br>{lambda()#3}::operator()() const:: {lambda(long)#1}>::result_type*) [clone .kd] | 42 /<br>42              | 0.010240            | 0.014853                       |
| unattributed / Other GPU bookkeeping<br>void (anonymous namespace)::softmax_warp_forward<float, float, float, 6, false,<br>false, 32>(float*, float const*, int, int, int, bool const*, int, bool) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 6 / 6                   | 0.002226            | 0.002352                       |
| unattributed / Other GPU bookkeeping<br>void at::native::_scatter_gather_elementwise_kernel<256, 4,<br>at::native::_cuda_scatter_gather_internal_kernel<false, at::native::OpaqueType<4>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}>(int,<br>at::native::_cuda_scatter_gather_internal_kernel<false, at::native::OpaqueType<4>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}) [clone .kd]                                                                                                                                                                                 | 48 /<br>48              | 0.015452            | 0.015559                       |
| unattributed / Other GPU bookkeeping<br>void at::native::_scatter_gather_elementwise_kernel<256, 4,<br>at::native::_cuda_scatter_gather_internal_kernel<true, at::native::OpaqueType<4>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}>(int,<br>at::native::_cuda_scatter_gather_internal_kernel<true, at::native::OpaqueType<4>,<br>long>::operator()<at::native::TensorAssign>(at::TensorIterator&, long, long, long,<br>at::native::TensorAssign const&)::{lambda(int)#1}) [clone .kd]                                                                                                                                                                                   | 6 / 6                   | 0.002579            | 0.003046                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
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| unattributed / Other GPU bookkeeping<br>void at::native::elementwise_kernel_manual_unroll<128, 4,<br>at::native::gpu_kernel_impl<at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::lambda()#3)::operator()() const::{lambda()#4)::operator()() const::{lambda(long)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>lambda()#3)::operator()() const::{lambda()#4)::operator()() const::{lambda(long)#1} const&>::lambda(int, bool)#1>(int,<br>at::native::gpu_kernel_impl<at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::lambda()#3)::operator()() const::{lambda()#4)::operator()() const::{lambda(long)#1}>(at::TensorIteratorBase&,<br>at::native::direct_copy_kernel_cuda(at::TensorIteratorBase&)::<br>lambda()#3)::operator()() const::{lambda()#4)::operator()() const::{lambda(long)#1} const&>::lambda(int, bool)#1)) [clone .kd]                                                                                                                                                                    | 48 /<br>48              | 0.020612            | 0.020886                       |
| unattributed / Other GPU bookkeeping<br>void at::native::elementwise_kernel_manual_unroll<128, 4,<br>at::native::gpu_kernel_impl_nocast<at::native::CUDataFunctor_add<int><br>>(at::TensorIteratorBase&, at::native::CUDataFunctor_add<int> const&)::lambda(int, bool)#1>(int, at::native::gpu_kernel_impl_nocast<at::native::CUDataFunctor_add<int><br>>(at::TensorIteratorBase&, at::native::CUDataFunctor_add<int> const&)::lambda(int, bool)#1)) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 42 /<br>42              | 0.021400            | 0.018933                       |
| unattributed / Other GPU bookkeeping<br>void at::native::elementwise_kernel_manual_unroll<128, 8,<br>at::native::gpu_kernel_impl_nocast<at::native::(anonymous namespace)::CompareFunctor<float> >(at::TensorIteratorBase&, at::native::(anonymous namespace)::CompareFunctor<float> const&)::lambda(int, bool)#1>(int,<br>at::native::gpu_kernel_impl_nocast<at::native::(anonymous namespace)::CompareFunctor<float> >(at::TensorIteratorBase&, at::native::(anonymous namespace)::CompareFunctor<float> const&)::lambda(int, bool)#1)) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 18 /<br>18              | 0.018944            | 0.019577                       |
| unattributed / Other GPU bookkeeping<br>void at::native::index_elementwise_kernel<128, 4,<br>at::native::gpu_index_kernel<at::native::index_kernel_impl<at::native::OpaqueType<4><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>, at::native::index_kernel_impl<at::native::OpaqueType<4><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1} const&, bool)::lambda(int)#1>(long,<br>at::native::gpu_index_kernel<at::native::index_kernel_impl<at::native::OpaqueType<4><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>, at::native::index_kernel_impl<at::native::OpaqueType<4><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1} const&, bool)::lambda(int)#1)) [clone .kd] | 109 /<br>109            | 0.053089            | 0.053589                       |
| unattributed / Other GPU bookkeeping<br>void at::native::index_elementwise_kernel<128, 4,<br>at::native::gpu_index_kernel<at::native::index_kernel_impl<at::native::OpaqueType<8><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>, at::native::index_kernel_impl<at::native::OpaqueType<8><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1} const&, bool)::lambda(int)#1>(long,<br>at::native::gpu_index_kernel<at::native::index_kernel_impl<at::native::OpaqueType<8><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1>(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>, at::native::index_kernel_impl<at::native::OpaqueType<8><br>> >(at::TensorIteratorBase&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>lambda(char*, char const*, long)#1} const&, bool)::lambda(int)#1)) [clone .kd] | 12 /<br>12              | 0.005945            | 0.005985                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------|--------------------------------|
| unattributed / Other GPU bookkeeping<br>void at::native::index_elementwise_kernel<128, 4,<br>at::native::gpu_index_kernel<at::native::index_put_kernel_impl<at::native::OpaqueTy<br>pe<8> >(at::TensorIterator&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>{lambda(char*, char const*, long)#1}>(at::TensorIteratorBase&, c10::ArrayRef<long>,<br>c10::ArrayRef<long>, at::native::index_put_kernel_impl<at::native::OpaqueType<8><br>>(at::TensorIterator&, c10::ArrayRef<long>, c10::ArrayRef<long>)::{lambda(char*,<br>char const*, long)#1} const&, bool)::{lambda(int)#1}>(long,<br>at::native::gpu_index_kernel<at::native::index_put_kernel_impl<at::native::OpaqueTy<br>pe<8> >(at::TensorIterator&, c10::ArrayRef<long>, c10::ArrayRef<long>)::<br>{lambda(char*, char const*, long)#1}>(at::TensorIteratorBase&, c10::ArrayRef<long>,<br>c10::ArrayRef<long>, at::native::index_put_kernel_impl<at::native::OpaqueType<8><br>>(at::TensorIterator&, c10::ArrayRef<long>, c10::ArrayRef<long>)::{lambda(char*,<br>char const*, long)#1} const&, bool)::{lambda(int)#1}) [clone .kd] | 6 / 6                   | 0.003059            | 0.002979                       |
| unattributed / Other GPU bookkeeping<br>void at::native::mbtopk::computeBlockDigitCounts<float, unsigned int, unsigned int,<br>2>(at::cuda::detail::TensorInfo<float const, unsigned int>, unsigned int, unsigned<br>int*, unsigned int, unsigned int, int, int, unsigned int, unsigned int, unsigned<br>int*, short*) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 24 /<br>24              | 0.050263            | 0.052110                       |
| unattributed / Other GPU bookkeeping<br>void at::native::mbtopk::computeBlockwiseWithinKCounts<unsigned int,<br>float>(unsigned int*, short*, unsigned int*, unsigned int, int, bool, unsigned<br>int*, float*, unsigned int*, unsigned int*, unsigned int*, unsigned int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24 /<br>24              | 0.037523            | 0.038716                       |
| unattributed / Other GPU bookkeeping<br>void at::native::mbtopk::fill<unsigned int, unsigned int>(unsigned int*, unsigned<br>int, unsigned int) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 / 6                   | 0.001612            | 0.001706                       |
| unattributed / Other GPU bookkeeping<br>void at::native::mbtopk::gatherTopK<float, unsigned int,<br>2>(at::cuda::detail::TensorInfo<float const, unsigned int>, unsigned int, unsigned<br>int, bool, unsigned int, unsigned int, at::cuda::detail::TensorInfo<float, unsigned<br>int>, unsigned int, at::cuda::detail::TensorInfo<long, unsigned int>, unsigned int,<br>unsigned int, unsigned int, float*, unsigned int*, unsigned int*, unsigned int)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6 / 6                   | 0.026179            | 0.026026                       |
| unattributed / Other GPU bookkeeping<br>void at::native::tensor_kernel_scan_innermost_dim<float, std::plus<float> >(float*,<br>float const*, unsigned int, unsigned int, unsigned int, float, std::plus<float>)<br>[clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6 / 6                   | 0.002646            | 0.002726                       |
| unattributed / Other GPU bookkeeping<br>void at::native::unrolled_elementwise_kernel<at::native::CUDAFunction_add<int>,<br>std::array<char*, 3ul>, 4, TrivialOffsetCalculator<2, unsigned int>,<br>TrivialOffsetCalculator<1, unsigned int>, at::native::memory::LoadWithoutCast,<br>at::native::memory::StoreWithoutCast>(int, at::native::CUDAFunction_add<int>,<br>std::array<char*, 3ul>, TrivialOffsetCalculator<2, unsigned int>,<br>TrivialOffsetCalculator<1, unsigned int>, at::native::memory::LoadWithoutCast,<br>at::native::memory::StoreWithoutCast) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 42 /<br>42              | 0.012020            | 0.012020                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<16, at::native::BinaryFunction<bool,<br>bool, bool, at::native::BitwiseOrFunction<bool> >, std::array<char*, 3ul> >(int,<br>at::native::BinaryFunction<bool, bool, bool, at::native::BitwiseOrFunction<bool> >,<br>std::array<char*, 3ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6 / 6                   | 0.002759            | 0.002659                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<16, at::native::FillFunction<bool>,<br>std::array<char*, 1ul> >(int, at::native::FillFunction<bool>, std::array<char*,<br>1ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 7 / 7                   | 0.002119            | 0.002185                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<16,<br>at::native::bitwise_not_kernel_cuda(at::TensorIteratorBase&)::{lambda(bool)#1},<br>std::array<char*, 2ul> >(int,<br>at::native::bitwise_not_kernel_cuda(at::TensorIteratorBase&)::{lambda(bool)#1},<br>std::array<char*, 2ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6 / 6                   | 0.002286            | 0.002332                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Calls<br>old /<br>fixed | Old ms<br>per round | Final fixed<br>ms per<br>round |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------|--------------------------------|
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::(anonymous namespace)::launch_clamp_scalar(at::TensorIteratorBase&, c10::Scalar, c10::Scalar, at::native::detail::ClampLimits)::{lambda()}#1::operator()() const:: {lambda()}#3::operator()() const:: {lambda(int)#1}, std::array<char*, 2ul> >(int, at::native::(anonymous namespace)::launch_clamp_scalar(at::TensorIteratorBase&, c10::Scalar, c10::Scalar, at::native::detail::ClampLimits):: {lambda()}#1::operator()() const:: {lambda()}#3::operator()() const:: {lambda(int)#1}, std::array<char*, 2ul>) [clone .kd]   | 42 /<br>42              | 0.012046            | 0.014173                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::(anonymous namespace)::launch_clamp_scalar(at::TensorIteratorBase&, c10::Scalar, c10::Scalar, at::native::detail::ClampLimits)::{lambda()}#1::operator()() const:: {lambda()}#4::operator()() const:: {lambda(long)#1}, std::array<char*, 2ul> >(int, at::native::(anonymous namespace)::launch_clamp_scalar(at::TensorIteratorBase&, c10::Scalar, c10::Scalar, at::native::detail::ClampLimits):: {lambda()}#1::operator()() const:: {lambda()}#4::operator()() const:: {lambda(long)#1}, std::array<char*, 2ul>) [clone .kd] | 6 / 6                   | 0.001772            | 0.001859                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::(anonymous namespace)::masked_fill_kernel(at::TensorIterator&, c10::Scalar const&):: {lambda()}#1::operator()() const:: {lambda()}#7::operator()() const:: {lambda(float, bool)#1}, std::array<char*, 3ul> >(int, at::native::(anonymous namespace)::masked_fill_kernel(at::TensorIterator&, c10::Scalar const&):: {lambda()}#1::operator()() const:: {lambda()}#7::operator()() const:: {lambda(float, bool)#1}, std::array<char*, 3ul>) [clone .kd]                                                                          | 6 / 6                   | 0.010073            | 0.010066                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::(anonymous namespace)::where_kernel_impl(at::TensorIterator&)::{lambda()}#1::operator()() const:: {lambda()}#11::operator()() const:: {lambda(bool, float, float)#1}, std::array<char*, 4ul> >(int, at::native::(anonymous namespace)::where_kernel_impl(at::TensorIterator&)::{lambda()}#1::operator()() const:: {lambda()}#11::operator()() const:: {lambda(bool, float, float)#1}, std::array<char*, 4ul>) [clone .kd]                                                                                                      | 12 /<br>12              | 0.004605            | 0.004878                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::BUnaryFunctor<int, int, int, at::native::binary_internal::div_floor_kernel_cuda(at::TensorIteratorBase&):: {lambda()}#1::operator()() const:: {lambda()}#3::operator()() const:: {lambda(int, int)#1}>, std::array<char*, 2ul> >(int, at::native::BUnaryFunctor<int, int, int, at::native::binary_internal::div_floor_kernel_cuda(at::TensorIteratorBase&):: {lambda()}#1::operator()() const:: {lambda()}#3::operator()() const:: {lambda(int, int)#1}>, std::array<char*, 2ul>) [clone .kd]                                  | 42 /<br>42              | 0.019407            | 0.015160                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::CUDataFunctorOnSelf_add<int>, std::array<char*, 2ul> >(int, at::native::CUDataFunctorOnSelf_add<int>, std::array<char*, 2ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                      | 48 /<br>48              | 0.013725            | 0.016085                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::CUDataFunctorOnSelf_add<long>, std::array<char*, 2ul> >(int, at::native::CUDataFunctorOnSelf_add<long>, std::array<char*, 2ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                    | 6 / 6                   | 0.001966            | 0.001899                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::CUDataFunctor_add<float>, std::array<char*, 3ul> >(int, at::native::CUDataFunctor_add<float>, std::array<char*, 3ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                              | 6 / 6                   | 0.001939            | 0.001992                       |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::FillFunctor<float>, std::array<char*, 1ul> >(int, at::native::FillFunctor<float>, std::array<char*, 1ul>) [clone .kd]                                                                                                                                                                                                                                                                                                                                                                                                          | 12 /<br>12              | 0.003618            | 0.003111                       |

| Scope / stage / compiled kernel                                                                                                                                                                                                                                                                                                                                              | Calls old / fixed | Old ms per round | Final fixed ms per round |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|--------------------------|
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::native::FillFunctor<int>, std::array<char*, 1ul>) [clone .kd]                                                                                                                                                      | 7 / 7             | 0.001932         | 0.001886                 |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_elementwise_kernel<4, at::native::bfloat16tofloat32_copy_kernel_cuda(at::TensorIteratorBase&):: {lambda(c10::BFloat16)#1}, std::array<char*, 2ul> >(int, at::native::bfloat16tofloat32_copy_kernel_cuda(at::TensorIteratorBase&):: {lambda(c10::BFloat16)#1}, std::array<char*, 2ul>) [clone .kd]        | 6 / 6             | 0.010886         | 0.009939                 |
| unattributed / Other GPU bookkeeping<br>void at::native::vectorized_gather_kernel<16, long>(char*, char*, long*, int, long, long, long, long, bool) [clone .kd]                                                                                                                                                                                                              | 6 / 6             | 0.002086         | 0.002252                 |
| unattributed / Other GPU bookkeeping<br>void at::native::warpMergeSortKVInPlace<2, -1, 128, 16, float, long, at::native::GTOp<float, true>, unsigned int, 32>(at::cuda::detail::TensorInfo<float, unsigned int>, unsigned int, unsigned int, unsigned int, at::cuda::detail::TensorInfo<long, unsigned int>, unsigned int, at::native::GTOp<float, true>, float) [clone .kd] | 6 / 6             | 0.004832         | 0.004999                 |

Prefill is outside this steady-decode timing table and has a separate causal/chunk-partition repair. CPU scheduling has no GPU-kernel duration here. The last row retains GPU bookkeeping outside the attributed model scopes; its individual kernels are listed without inventing missing semantic labels.

## 5. Brief unprofiled engine controls — 60K input, not 60K output

The approximately 80–87 tok/s figures are short controls on one 60,000-input- token Pi fixture. They are not results from generating 60,000 tokens.

| Compiled configuration           | Natural responses | Output tokens | Timed post-first seconds | Median round | Pooled post-first rate |
|----------------------------------|-------------------|---------------|--------------------------|--------------|------------------------|
| Original compiled M8             | 3                 | 2,582         | 30.142                   | 59.684 ms    | 85.561 tok/s           |
| Final compiled M8: Fix 1 + Fix 2 | 3                 | 2,525         | 29.902                   | 60.052 ms    | 84.342 tok/s           |

These fresh controls use the same builds as the stage profiles, in separate unprofiled passes. All three natural responses per arm are retained in the [profile and control audit](#).

These direct-engine controls exclude HTTP/Pi, tools, snapshot publication, cold prefill and warm-up. Rates pool tokens and time after the first output chunk; round latency is the median of the per-response steady-round medians. Throughput also depends on speculative acceptance, so it is not a pure kernel speed ratio. Clean timing passes do not record GPU profiles or copy activations.

### Cost of preserving compiled BF16 rounding

Four fresh compiled runs used **before–after–after–before order**, with three natural responses per run on the same 60,000-token Pi prefix. Both settings use the final four performance repairs, full BF16 target head and piecewise GPU graphs. The only changed compiler setting is TORCHINDUCTOR\_EMULATE\_PRECISION\_CASTS=0/1. The stage profiles in section 4 now include this rounding alignment. The following separate ABBA comparison isolates Fix 2 from Fix 1.

| Compiled setting                  | Natural responses | Output tokens | Median round | Committed tokens/round | Pooled post-first rate |
|-----------------------------------|-------------------|---------------|--------------|------------------------|------------------------|
| Before rounding alignment         | 6                 | 4,474         | 60.906 ms    | 5.288                  | 84.123 tok/s           |
| BF16 intermediate casts preserved | 6                 | 5,050         | 60.900 ms    | 5.211                  | 83.070 tok/s           |

Preserving casts changed the median round by **-0.006 ms (-0.01%)** and the pooled token rate by **-1.25%**. Mean committed tokens per round changed by -1.45%. The per-response median rounds ranged from 60.245 to 61.415 ms before and 60.567 to 61.194 ms after.

Round time is effectively unchanged in this sample. The slightly lower token rate is consistent with fewer committed tokens per round. Every run used the same sampling settings and three seeds, but changed rounding produced different natural continuations. The nearest-even native RoPE correction applies to the **eager reference**; this measures the compiled half of the alignment. Startup, compilation, warm-up and cold prefill are excluded from the decode timings. All twelve natural completions are included; per-round timings and source identities are retained in [rounding-speed-abba.json](#).

## Separate, earlier 60K-generated-token verify-head experiment

This earlier compiled experiment used 115 natural responses per method across 11 intact Pi request boundaries, with 57,008–65,527 input tokens. It predates the final D7 repairs and **does not measure the final corrected M8's long-run speed**. The head methods also have different numerical guarantees.

| Earlier head method           | Natural responses | Generated tokens | Timed post-first seconds | Post-first tok/s |
|-------------------------------|-------------------|------------------|--------------------------|------------------|
| Full BF16 reference head      | 115               | 60,598           | 946.230                  | 63.920           |
| INT2 block-8 / rerank-80      | 115               | 60,075           | 892.548                  | 67.178           |
| INT2 global-128 / BF16 rerank | 115               | 60,348           | 892.266                  | 67.506           |
| INT2 global-256 / BF16 rerank | 115               | 60,675           | 906.873                  | 66.779           |

The first output chunks are excluded from both the rate numerator and timed denominator. See [earlier-60k-output-speed.json](#).

## 6. Repairs and upstream ownership

1. **GDN prefill**: raw per-token gates and an ordered transition avoid the diagnosed dependence on future gates, cumulative-gate rounding and chunk partition. Core implementation belongs with libr4d; adapters belong in Radiance.
2. **GDN decode**: match serial convolution arithmetic, gate precision, recurrent reductions, output rounding and retained-prefix state. Rejecting a speculative suffix must not make its state persistent. Packed QKV views remove copies.
3. **Gated normalization**: retain the serial row tile without serializing all eight rows. The vendored FLA dispatcher belongs in vLLM.
4. **Attention**: retain each query's causal range and serial reduction/split policy; share KV reads to recover performance. Core ownership is libr4d.
5. **Residual/QK/final normalization**: preserve the chosen finite-precision boundaries; retain FP32 residual sums in registers. Existing vLLM rounding PRs must be checked for overlap before proposing another repair.
6. **Full BF16 head**: interleave two arithmetic-preserving M4 groups in one HIP launch. This avoids the first repair's duplicated launch/concatenation overhead. The upstream skinny-GEMM implementation belongs in vLLM.
7. **Fix 1 compiled integration**: install dispatch before tracing/capture, use opaque bindings where necessary, and validate startup-only fallbacks separately from real requests. Integration and conformance adapters belong in Radiance.
8. **Fix 2, execution-mode rounding**: enable TORCHINDUCTOR\_EMULATE\_PRECISION\_CASTS=1 for the compiled target and preserve nearest-even BF16 products in native eager MROPE. These changes address a separate eager/compiled discrepancy after Fix 1 had aligned M1 and M8.

These are submission scopes, not a claim that the pinned adapters apply unchanged to current upstream main. The new ports require their own tests. Upstream links and submission state are maintained in [submissions.md](#).

## 7. Execution-mode diagnosis and remaining limits

The eager/compiled mismatch was localized to intermediate BF16 casts and native RoPE multiplication. Aligning both produced exact eager/compiled **M8** output agreement on the controlled 320-position Pi replay. This is separate from the fresh 10K eager-M1/compiled-M8 before/after comparison in section 3, which measures both fixes together across all 23 continuations.

The [complete boundary comparison](#) also matches captured inputs and outputs at all 465 matched boundaries, across 320 decode and nine sampled prefill positions. This sampled activation check supplements the whole-model and isolated-stage results; it does not cover all arbitrary inputs or every persistent state transition.

## Intermediate BF16 casts

Before alignment, the first difference occurred in layer 0's MLP SiLU/gating, after identical GDN and gate/up projection outputs. The [isolated native replay](#) tested all 64 layers on common inputs: 20,480 decode layer-position evaluations and 576 sampled prefill evaluations. Each kernel reproduced its own captured output; none of the complete decode output rows agreed between modes.

```
eager:    BF16(BF16(SiLU(gate)) * up)
compiled: BF16(SiLU_FP32(gate) * up)
```

The BF16-intermediate arithmetic oracle matched eager in all 20,480 decode evaluations. An independent FP32-intermediate calculation still differed from the generated compiled kernel at 17 individual values, so that formula alone is not a bit-exact specification of the compiled operation.

Attention sigmoid gating had the same cast-elision distinction: eager rounded the sigmoid to BF16 before multiplying; compiled retained it in FP32. A [common-input native pilot](#) reproduced each implementation's own output and found 0/8 equal output rows between modes.

TORCHINDUCTOR\_EMULATE\_PRECISION\_CASTS=1 preserves these intermediate casts. The experiment checks the loaded compiler boolean as well as the environment setting. Cast preservation alone did not restore equality, in the [cast-only control](#); it exposed the remaining RoPE difference after matching the first three layers.

## Native RoPE multiplication

With compiler casts preserved, the first remaining discrepancy occurred in layer 3 after identical QKV projection and Q/K normalization. The [fine attention comparison](#) found:

| Layer 3 boundary       | Exact decode rows / 320 | Exact sampled prefill rows / 9 | Differing decode values |
|------------------------|-------------------------|--------------------------------|-------------------------|
| QKV projection         | 320                     | 9                              | 0                       |
| Query normalization    | 320                     | 9                              | 0                       |
| Key normalization      | 320                     | 9                              | 0                       |
| Query after RoPE       | 0                       | 1                              | 184,437                 |
| Key after RoPE         | 0                       | 1                              | 30,984                  |
| Value input            | 320                     | 9                              | 0                       |
| Attention output       | 0                       | 1                              | 1,335,594               |
| Gate input             | 320                     | 9                              | 0                       |
| Gated attention output | 0                       | 1                              | 1,282,380               |

The later output differences inherit changed rotary inputs and state; this table does not establish separate attention-kernel defects. On the same normalized Q/K and selected BF16 cosine/sine coefficients, the [explicit arithmetic oracle](#) reproduced each mode exactly at all 320 decode and nine sampled prefill positions:

```
native eager:    RNE_BF16(RTZ_BF16(a*c) - RTZ_BF16(b*s))
compiled emulation: RNE_BF16(RNE_BF16(a*c) - RNE_BF16(b*s))
```

The second rotary half uses addition with the same product-rounding rules; the 192 non-rotary coordinates are unchanged. RTZ means round toward zero; RNE means round to nearest, ties to even. This isolates rotation arithmetic, not the separate selection of cosine/sine coefficients.

The [native replay](#) then compared the installed kernel with a private copy that changed only product rounding:

| Native RoPE variant            | Q and K exact versus original eager | Q and K exact versus compiled cast preservation |
|--------------------------------|-------------------------------------|-------------------------------------------------|
| Original native multiplication | 320/320 decode; 9/9 prefill         | 0/320 decode; 1/9 prefill                       |

| Native RoPE variant                  | Q and K exact versus original eager | Q and K exact versus compiled cast preservation |
|--------------------------------------|-------------------------------------|-------------------------------------------------|
| Explicit nearest-even multiplication | 0/320 decode; 1/9 prefill           | 320/320 decode; 9/9 prefill                     |

Both Q and K independently meet those counts. Position zero is the identity rotation. Input/coefficient immutability and injected one-bit errors are checked. Source, LLVM IR, AMD assembly and GPU-binary hashes bind the result to the executed implementations; the full-model worker also verifies that the modified kernel runs during the request.

The installed **Triton 3.7.1 / PyTorch 2.12.0+rocm7.14** build lowers BF16 multiplication to `llvm.amdgcn.fdot2.bf16.bf16`, emitted as `v_dot2_bf16_bf16` on the R9700. Its observed truncation is the defect already fixed by [Triton PR #11227](#), merged August 14, 2026. That fix replaces the special lowering with an FP32 multiply and explicit nearest-even conversion. The new evidence here is the defect's occurrence and causal isolation inside the captured Qwen RoPE path; the upstream repair is credited to its existing authors.

## Remaining coverage limits

- The fresh 10K comparison measures eager M1 versus compiled M8 before and after both fixes. The separate eager-M8/compiled-M8 comparison covers 320 positions. Neither establishes arbitrary-input equality. The brief compiled speed comparison is reported in §5; instrumented replay durations are not serving speed.
- Captured activation equality does not certify uncaptured KV/GDN/convolution state, unpaired operations or independent MRoPE coefficient selection.
- The 10K replays cover the all-seven-accepted D7 path. Small operator tests exercise rejection boundaries and injected faults, but full model coverage of every rejection history, long-context range, concurrency, snapshot restore and hardware platform is not established.
- Neither experiment proves arbitrary-input equivalence to an independent mathematical reference, eliminates model-generated loops or measures improved coding accuracy. Evidence remains tied to its source, binary and configuration identities; changed implementations require qualification.

## 8. Evidence and reproduction

The `evidence/` directory contains aggregate comparisons and kernel profiles, with file SHA-256 values in `evidence-sha256.json`. Recompute the stage table, kernel accounting and aggregate assertions without GPU use:

```
uv run python reports/d7-rdna4-2026-09-17/build_report.py
```

The original Pi transcripts, token IDs, ranked token IDs and raw traces remain private. The aggregate measurements can be audited from the published receipts; independent end-to-end reproduction needs a public replacement workload. Synthetic operator regressions accompany the corresponding source submissions.

The [10K before/after audit](#) binds all four runs to the same corpus and records their source, repair, compiler, graph and per-continuation receipt identities. The final eager M1 reference and final compiled M8 candidate both include Fix 1 and their respective Fix 2 rounding repairs.

## References and attribution

- [Radiance](#), the qualified downstream stack.
- [libr4d](#), RDNA4 attention and recurrent kernels.
- [vLLM](#), including ROCm skinny GEMMs and vendored FLA.
- [Flash Linear Attention](#), serial GDN/normalization foundations.
- [vLLM batch invariance](#), related existing work.
- [PyTorch numerical accuracy](#), batched/reduction-order limitations.
- Existing normalization work: [vLLM #49639](#) and [#52243](#).

This contribution builds on those implementations. Batch invariance itself is not introduced by this report. Earlier DFlash RNG work in Radiance PR #8 backports [vLLM #54282](#) and is not presented as a new upstream discovery here.